



Working Paper

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The Composition of Federal R&D Spending and U.S. Productivity Growth

WR-A4392-1

August 2026

Supported by the Diller-von Furstenberg Family Foundation and Pershing Square Philanthropies

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Published by the RAND Corporation, Santa Monica, Calif.

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About This Working Paper

Recent research suggests that the composition of public research and development (R&D) investment such as for basic versus applied research or for plant R&D—not just the overall amount spent on R&D—matters for productivity outcomes. We present new, corroborative evidence using a local projections method to estimate dynamic responses of macroeconomic productivity outcomes to unanticipated changes in R&D composition. We find that if R&D spending shifted toward basic research (i.e., research to expand the knowledge base without specific applications) are associated with the largest cumulative productivity gains over a ten-year horizon, operating primarily through total factor productivity. By contrast, spending shifts toward development activities (i.e., using research to develop or improve products and systems) are associated with cumulative productivity declines of comparable magnitude. This suggests that the current development share exceeds its productivity-maximizing level. We then use an illustrative portfolio analysis to benchmark the historical federal R&D portfolio against alternative allocations, finding that Pareto-optimal portfolios place roughly three times the historical share on basic research and substantially lower shares on development. These results have practical significance for science and technology policy, particularly in the era of increasing budget tradeoffs against R&D spending. This report is written primarily for economists working on innovation, productivity, and science policy. The report may also interest policy analysts engaged with federal R&D spending, though the analysis presumes familiarity with applied macroeconometric methods.

RAND von Furstenberg Family Budget Model Initiative

This work was conducted as part of the RAND von Furstenberg Family Budget Model Initiative, which developed tools for analysis of the U.S. tax code and federal spending and for forecasting the effects of proposed legislation. The RAND von Furstenberg Family Budget Model Initiative operates within RAND Education, Employment, and Infrastructure, a division of RAND that aims to improve educational opportunity, economic prosperity, and civic life for all. For more information, visit www.rand.org/budget-model or email EEI@rand.org.

Funding

The RAND von Furstenberg Family Budget Model Initiative is generously supported by the Diller-von Furstenberg Family Foundation and Pershing Square Philanthropies. Initial funding for the budget model was provided by the Nick and Leslie Hanauer Foundation.

Acknowledgments

The authors thank the Budget Model team, and especially Carter Price, Aaron Strong, Akshaya Suresh, and Jeffrey Wenger. Carter Price generously reviewed an early draft of this piece, and his comments and support were instrumental. The report's peer reviewer, Vegard Nygaard, provided a thorough and insightful review, which improved upon the piece. We also appreciate thoughtful comments and input from Stephanie Rennane. We also appreciate the work of Mirka Vuollo for coordinating the quality assurance process.

Summary

- Different categories of federal R&D show different associations with U.S. productivity growth.
 - A one-standard-deviation increase in the share of basic research is associated with cumulative labor productivity gains of about 2.1 percentage points over ten years.
 - A comparable increase in the development R&D share is associated with cumulative declines of about 2.2 percentage points.
 - Applied research and R&D plant investments are associated with positive but less precisely estimated effects of 1.7 and 1.4 percentage points.
- The productivity effects of basic research operate primarily through total factor productivity (roughly three-quarters of the cumulative effect). By contrast, for R&D plant, capital deepening is the only positive channel that is statistically significant. The mechanisms for applied research effects are split between the two.
- Holding fixed total research and development spending, our illustrative portfolio optimization framework, which treats R&D categories as productivity investments, results in Pareto-optimal compositions that allocate roughly three times the historical share to basic research (47 percent vs. 17 percent) and reduce by a factor of three the historical share on development research (18 percent vs 59 percent). We note that this framework solely optimizes for productivity, thus not accounting for other potential research objectives (e.g., military competitiveness).

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The Composition of Federal R&D Spending and U.S. Productivity Growth

The United States federal government has invested in research and development for decades, with annual expenditures recently estimated to be roughly \$200 billion (NCES, Survey of Federal Funds for Research and Development, July 2025).¹ This investment spans a diverse portfolio of activities, from fundamental scientific research with no immediate commercial application to the development of specific technologies for defense, health, and other missions. A substantial literature documents that aggregate R&D investment generates positive returns to productivity growth (Gullo, et. al. 2025). Yet relatively little attention has been paid to whether the composition of this investment matters for productivity outcomes, a question of considerable policy relevance.

This compositional question has practical significance for science and technology policy, particularly in an era of constrained federal spending on R&D. If different types of federal R&D are associated with substantially different productivity effects, then reallocating resources toward higher-productivity federal R&D types could achieve more productivity growth from the same total spending. Because productivity growth is one of several legitimate objectives of federal R&D spending, we do not claim that this reallocation is strictly optimal. Our analysis focuses on productivity outcomes and does not address how the composition might be evaluated against other objectives such as national security, public health, human capital development, or basic scientific capacity.

This work considers four types of R&D spending: basic research, applied research, development, and research infrastructure (“plant”) investment. Distinctions among these types reflect genuine differences in the nature of scientific activity, its time horizons, and its channels of influence on economic outcomes.

- Basic research aims to expand fundamental knowledge without specific applications in mind, potentially generating broad spillovers across industries and over extended time periods. Examples include: decoding brain activity related to speech², understanding

¹ Vannevar Bush, *Science, the Endless Frontier: 75th Anniversary Edition*, National Science Foundation, 2020 (report to the President originally submitted July 1945). As of July 31, 2026: https://nsf.gov/resources/nsf.gov/2023-04/EndlessFrontier75th_w.pdf. See also National Science Board, “Federal Support for U.S. R&D,” in *Discovery: R&D Activity and Research Publications*, Science and Engineering Indicators, NSB-2025-7, July 23, 2025. As of July 31, 2026: <https://nces.nsf.gov/pubs/nsb20257/federal-support-for-u-s-r-d>

² National Institutes of Health, “Scientists Create Speech Using Brain Signals,” *NIH Research Matters*, May 7, 2019. As of July 31, 2026: <https://www.nih.gov/news-events/nih-research-matters/scientists-create-speech-using-brain-signals>

materials related to quantum computing³, and identifying the patterns of agricultural pests⁴. As of 2023, basic research comprised roughly 25 percent of total federal R&D annual expenditures.⁵

- Applied research directs scientific effort toward practical objectives, possibly offering more immediate but narrower benefits. Examples include: navigational technologies that became the basis for Global Positioning System (GPS)⁶ and novel pharmaceuticals for rare diseases.⁷⁸ As of 2023, applied research comprised roughly 29 percent of total federal R&D annual expenditures.
- Development activities translate research findings into creating or improving usable products and systems. Examples include: hypersonic weapons systems⁹ and nuclear energy technologies¹⁰. Development activities currently comprises roughly 45 percent of total federal R&D annual expenditures.
- Research plant and equipment investments, such as laboratory construction and specialized instrumentation, create the physical infrastructure that enables scientific activity, representing a form of capital deepening specific to the innovation sector.¹¹ Examples include: national labs and infrastructure such as particle colliders and supercomputers. As of 2023, research plant & equipment investments equate to about 3% (roughly \$5.7 billion) of federal R&D annual expenditures, though this category of funding is accounted for separately.

This paper provides new evidence on the association between productivity and federal R&D composition using a reduced-form empirical approach. First, we document substantial

³ U.S. Department of Defense, Basic Research Office, “New Breakthrough in Quantum Engineering!” November 12, 2024. As of July 31, 2026: <https://basicresearch.defense.gov/News/Articles/News-Display/Article/3962163/new-breakthrough-in-quantum-engineering/>

⁴ Alan Burdick and Emily Anthes, “Nine Federally Funded Scientific Breakthroughs That Changed Everything,” *New York Times*, May 16, 2025. As of July 31, 2026: <https://www.nytimes.com/2025/05/16/science/federally-funded-science-breakthroughs.html>

⁵ Data on research spending come from the National Science Foundation, “Table DISC-8: Federal Obligations for R&D and R&D plant, by agency and performer: FY 2023,” National Center for Science and Engineering Statistics, Survey of Federal Funds for Research and Development, Vol 73, FYs 2023-2024, July 23, 2025.

⁶ Alan Burdick and Emily Anthes, “Nine Federally Funded Scientific Breakthroughs That Changed Everything,” *New York Times*, May 16, 2025. As of July 31, 2026: <https://www.nytimes.com/2025/05/16/science/federally-funded-science-breakthroughs.html>

⁷ U.S. Office of Management and Budget, “Department of Energy,” Performance.gov, Trump Administration archive, undated. As of July 31, 2026: <https://trumpadministration.archives.performance.gov/energy/>

⁸ American Association for the Advancement of Science, “Research on R&D Funding: The Different Functions of Public and Private R&D,” AAAS R&D Budget and Policy Program, September 2025. As of July 31, 2026: <https://www.aaas.org/news/research-rd-funding-different-functions-public-and-private-rd>

⁹ National Security Technology Accelerator, “U.S. Military R&D Trends for 2023,” April 2023. As of July 31, 2026: <https://nstxl.org/u-s-military-rd-trends-for-2023/>

¹⁰ U.S. Department of Energy, Office of Nuclear Energy, “Our Budget,” undated. As of July 31, 2026: <https://www.energy.gov/ne/our-budget>

¹¹ These definitions are consistent with the OMB Circular A-11, as annotated in National Center for Science and Engineering Statistics, “Definitions of Research and Development: An Annotated Compilation of Official Sources,” NCSES 22-209, May 19, 2022.

heterogeneity in the estimated productivity associations across types of R&D spending. This result indicates that, compared to the current development-heavy R&D spending mix, shifting dollars away from development and toward other categories of R&D would result in stronger productivity growth, all else equal. This reallocation should be read as a labor productivity result, not a complete accounting for all policy tradeoffs, such as defense capabilities, public health technologies, or scientific infrastructure.

Second, we find that these compositional associations substantially exceed the associations we can detect from positive shocks to overall R&D spending. When we examine shocks to total federal R&D without distinguishing composition, the estimated cumulative productivity response is positive but imprecisely estimated, with confidence intervals that include zero.

Third, we develop a portfolio optimization framework that treats different R&D types as assets with heterogeneous expected returns and uncertainties. Our analysis suggests that the historical composition of federal R&D has been substantially over-invested in development and thus undershooting the potential maximum productivity gains. Optimal portfolios in our framework, which seek to maximize long-run productivity gains, nearly triple the investment in basic research and cut investment in development by roughly 40 percentage points.

While such calculations require strong assumptions about the stability of estimated effects and the feasibility of reallocation, they provide a useful benchmark for evaluating the productivity implications of compositional choices. Our analysis characterizes associations with labor productivity in the U.S. private business sector. Thus, implications for reallocating funding away from development are therefore conditional on labor productivity being the relevant policy objective.

This analysis combines detailed data on federal R&D budget authority by function with the productivity accounts maintained by John Fernald at the Federal Reserve Bank of San Francisco (Fernald, 2014), constructing an annual panel spanning 1956 through 2022. This period encompasses substantial variation in both the level and composition of federal R&D, driven by the Cold War buildup, the Space Race, shifting defense priorities, the rise of health research funding, and periodic changes in science policy emphasis. Figures 1 and 2 in the next section present the evolution of R&D composition and productivity over this period.

Our empirical strategy employs Local Projections (LP) as introduced by Jordà (2005) to estimate dynamic responses for macroeconomic productivity outcomes to unanticipated shifts in R&D composition (referred to throughout this paper as “shocks”). The LP method accommodates straightforward computation of cumulative effects and permits flexible treatment of the relationship between short-run and long-run responses. We construct R&D composition shocks as residuals from forecasting equations that control for the predictable component of each series, aggregate R&D dynamics, and macroeconomic conditions. These innovations represent

the unpredictable component of compositional change, though we emphasize that unpredictability does not guarantee exogeneity with respect to productivity outcomes.¹²

We subject our findings to extensive robustness analysis. Diagnostic tests confirm that our constructed shocks exhibit the statistical properties expected of unpredictable shocks, passing standard whiteness tests and show minimal correlation with lagged predictors. Estimates remain qualitatively stable across alternative lag specifications, subsample periods, and inference methods. Lead tests designed to detect pre-existing trends yield mixed evidence, with most specifications showing insignificant lead coefficients but some marginal evidence of anticipation at longer horizons for certain shocks. We interpret this pattern as consistent with either gradual policy implementation or the presence of confounding factors that our approach cannot fully address. We evaluate the robustness of the precision of our estimates using a placebo test designed to shift the timing of R&D share innovations, to which our results remain strongly robust.

Our empirical strategy constructs shocks to R&D spending and funding composition as a forecast residual. Below, the robustness analysis narrows the set of stories under which our estimates could be biased. The whiteness diagnostic indicates that the shocks are not predictable from the information set used in the forecasting model, so any confounder operating through that information set is already absorbed into the forecast. Lead tests indicate that future R&D shocks generally do not predict current productivity outcomes, which speaks against confounders operating through pre-existing trends or reverse causality. Placebo timing tests confirm that the measured share-shock effects align with the actual timing of the constructed shocks. Together, these checks raise the bar for what an unobserved confounder would need to look like to drive our results.

The remainder of the report reviews related literature, describes the data and methods, presents the results and robustness analysis, and discusses the portfolio application and policy implications.

Prior Work Shows Heterogeneous R&D Returns; This Paper Quantifies the Composition Effect

The literature on R&D and productivity supports four claims that motivate our analysis:

¹² For instance, a new administration that simultaneously shifts the R&D portfolio toward basic research as part of a broader pro-growth agenda, which might include tax reform, deregulation, or trade liberalization, would generate composition innovation correlated with productivity gains through non-R&D channels. More subtly, if policymakers observe early signals of a productivity slowdown and respond by redirecting R&D toward applied research or development in an attempt to generate near-term results, the resulting composition innovation would reflect private information about future productivity that our forecasting equations cannot capture. We probe this concern using lead tests.

1. R&D investment generates positive productivity returns, with estimated social returns exceeding private returns by a factor of two or more (Griliches 1979; Jones and Williams 1998; Hall, Mairesse, and Mohnen 2010; Bloom, Schankerman, and Van Reenen 2013).
2. Federally funded R&D complements rather than crowds out private investment and is associated with larger productivity spillovers than privately funded R&D (Guellec and Van Pottelsberghe de la Potterie 2004; Aghion et al. 2008; Howell 2017; Azoulay et al. 2019; Moretti, Steinwender, and Van Reenen 2019).
3. The composition of R&D spending matters along the dimensions prior work has studied (e.g. public versus private, and defense versus nondefense) with differences in estimated spillovers across these categories (Corrado et al. 2023; Dyevre 2024; Fieldhouse and Mertens 2024).
4. Basic research is theorized and observed to expand the knowledge frontier with broad, slow-acting spillovers that differ in character from those generated by applied research and development (Arrow 1959; Nelson 1959; Mansfield 1980; Fleming and Sorenson 2004; Akcigit, Hanley, and Serrano-Velarde 2021).

Overall, the literature establishes that who funds R&D matters for productivity, but it is largely silent on how productivity outcomes vary with the composition of federal R&D across activity types — basic research, applied research, development, and R&D plant. Quantifying that composition effect is the contribution of this paper.

Detailed Discussion of Existing Literature

A large literature documents that R&D investment generates positive returns to productivity growth, with social returns typically estimated to exceed private returns by a factor of two or more (Griliches 1979; Jones and Williams 1998; Bloom, Schankerman, and Van Reenen, 2013; Hall, Mairesse, and Mohnen, 2010; Fieldhouse & Mertens 2024). Within this literature, which focuses on the US, a substantial strand examines public R&D specifically, generally finding that government-funded research complements rather than crowds out private efforts (Moretti, Steinwender, and Van Reenen, 2019; Guellec and Van Pottelsberghe de la Potterie, 2004; Azoulay et. al. 2019). Examples of mechanisms for “crowding-in” effect include different research incentives for publicly funded academia and the private sector (Aghion et. al. 2008), as well as the increase in venture funding for entrepreneurs that win early-stage R&D grants for small businesses (Howell 2017). In aggregate, public spending on R&D appears to lead to significant productivity impact through positive spillovers to innovation and economic opportunity.

Compositional studies tend to find that public and nondefense R&D spending have the greatest productivity spillovers, relative to private and defense R&D spending. Recent empirical work find that public R&D spending generates two to three times larger productivity spillovers, relative to private R&D spending (Dyevre 2024). The compositional shift from public to private space R&D may explain a recent finding of reduced space R&D spillovers over time (Carrado et.

al. 2023).¹³ As further evidence of compositional effects, nondefense R&D spending appears to offer very high social returns, relative to defense R&D (Dyevre 2024; Fieldhouse & Mertens 2024). Significant evidence suggests that the composition of R&D spending matters, both in terms of the source of the funding and the purpose of the project.

Theoretical work has emphasized that basic research may be particularly productive because it expands the frontier of applicable knowledge with broad spillovers that compound over time (Akcigit, Hanley, and Serrano-Velarde, 2021; Fleming and Sorenson, 2004; Nelson 1959; Arrow 1959).¹⁴ At the same time, some in the field have argued that the tradition of distinguishing between basic and other forms of research is arbitrary, with a “lack of a substantive basis” (Mowery & Rosenberg 1989). Yet past efforts to segment findings between basic and applied research suggest that basic research affects productivity with substantially longer lags than development (Mansfield, 1980). We accept that this distinction remains meaningful both in terms of how the government classifies R&D spending (OMB Circular A-11), as well as how research incentives operate (Aghion et. al. 2008).

Existing insights demonstrate the link between R&D spending and productivity and establish that not all R&D spending contributes equally. Yet these insights tend to focus on the deployer of research funds (either defense or nondefense, public or private sector), rather than the types of research activity being funded.

Our contribution is to estimate how the composition of federal R&D portfolio affects disaggregated components of U.S. productivity growth: total factor productivity, capital deepening, and labor productivity. Disaggregation of both R&D spending types and macroeconomic productivity allows us to characterize which categories of federal R&D are associated with gains in total factor productivity, capital deepening, or labor productivity, and with what magnitude and timing. This information matters for policy because the federal budget process involves explicit allocation decisions across these categories. If their productivity associations differ substantially, the composition of the portfolio is a policy lever in its own right. In this approach, we are contributing to a broader conversation on how policymakers can adjust their modeling approaches for the budgetary impacts of R&D spending. Currently, policy analysts do not integrate the existing research on R&D productivity spillovers, instead treating R&D investments as similar to infrastructure investments (Gullo, et. al. 2025).

¹³ For additional examples, Azoulay et. al. (2019) find that for every \$10 million in research funding from the National Institute of Health, there is an increase of 2.7 net patents in the pharmaceutical and biotechnology industries. Public research funding can also catalyze new enterprises. A quasi-experimental study found that an R&D subsidy from the U.S. Department of Energy’s Small Business Innovation Research program approximately doubles venture capital income, with large, positive impacts on patenting and revenue (Howell, 2017).

¹⁴ Aghion et. al. (2008) theorize that this difference in scope arises from different incentive structures: basic R&D spending tends to be more aligned to the academic research model, where researchers tend to have more creative control, ideal for exploratory research topics with broad applications. By contrast, applied R&D spending tends to be more aligned with private-sector focus on development, with a narrower set of applications.

Data Combine Public Sources on Productivity and R&D Spending

This section describes the sources and construction of our data, presents descriptive statistics, and discusses the historical variation in federal R&D composition that provides the foundation for our analysis.

Productivity Data

Our productivity measures come from the dataset maintained by John Fernald at the Federal Reserve Bank of San Francisco, which provides quarterly estimates of total factor productivity, labor productivity, and the components of labor productivity growth (Fernald, 2014). Fernald's methodology follows the growth accounting tradition pioneered by Solow (1957) and refined by Jorgenson, Gollop, and Fraumeni (1987), decomposing output growth into contributions from capital services, labor hours, labor quality, and a residual attributed to total factor productivity.

The growth accounting identity expresses labor productivity growth as the sum of three components.¹⁵ Total factor productivity captures improvements in the efficiency with which inputs are combined to produce output, representing technological change and other factors that raise output without requiring additional inputs. Capital deepening reflects increases in the capital-labor ratio, weighted by capital's share of income. Labor quality captures improvements in the composition of the workforce, including the effects of rising educational attainment and experience accumulation.

We use annual averages constructed from the quarterly data, covering the period 1956 through 2022. For our main specifications, we use the utilization-adjusted TFP measure, which attempts to purge measured productivity of cyclical variation in the intensity with which capital and labor are employed. This adjustment is important in our setting because federal R&D spending may respond to business cycle conditions that independently affect measured

¹⁵ The decomposition of labor productivity follows the standard growth accounting framework (Solow, 1975; Jorgenson, Gollop, and Fraumeni, 1987) as implemented by Fernald (2014). Under constant returns to scale and competitive factor markets, output growth in the business section can be written as

$$\Delta \ln Y = \alpha \Delta \ln K + (1 - \alpha) \Delta \ln L + \Delta \ln A$$

Where K is a composition-adjusted capital input, L is a composition-adjusted labor input, A is Hicks-neutral technology, and α is capital's share of income. Composition-adjusted labor input decomposes into raw hours H and labor quality Q

$$\Delta \ln L = \Delta \ln Q + \Delta \ln H.$$

Labor productivity is $LProd = \frac{Y}{H}$, so $\Delta \ln LProd = \Delta \ln Y - \Delta \ln H$. Substituting and rearranging gives the identity used in our analysis, which is a function of capital deepening $\left(\frac{K}{H}\right)$, labor quality (Q), and TFP (A)

$$\Delta \ln LProd = \alpha \Delta \ln \left(\frac{K}{H}\right) + (1 - \alpha) \Delta \ln Q + \Delta \ln A.$$

This decomposition relies on constant returns to scale, constant factor shares, and competitive factor markets, but our empirical results and substantive conclusions are independent of these assumptions. The decomposition serves as an accounting framework for tracing the channels through which R&D composition shocks operate.

productivity. All three components are measured in percentage points of annual growth and sum to labor productivity growth.

Federal R&D Data

Data on federal R&D spending by function come from historical tables published by the National Science Foundation and the Office of Management and Budget. These tables report budget authority for research and development activities, classified into four categories.¹⁶

- **Basic research** encompasses “systematic study directed toward greater knowledge or understanding of the fundamental aspects of phenomena,” without specific applications in view (Moris and Pece, 2022).
- **Applied research** involves “systematic study to understand the means to meet a recognized and specific need” (Moris and Pece, 2022).
- **Development** represents “systematic use of the knowledge and understanding gained from research directed toward the production of useful materials, devices, systems, or methods” (Moris and Pece, 2022).
- **R&D plant** includes construction of facilities and acquisition of equipment specifically for R&D activities (Moris and Pece, 2022).

We construct shares of each category in total federal R&D, which sum to one by definition. We also examine log levels of each category’s spending, which allows us to study the effects of increasing one type of R&D while controlling for others. We express all spending data in real terms using the GDP deflator.

Macroeconomic Controls

Our analysis includes two macroeconomic control variables that may affect both R&D composition decisions and productivity outcomes. The output gap, computed as the percentage deviation of real GDP from potential GDP, captures cyclical conditions that could influence both government spending priorities and measured productivity. The change in the federal funds rate controls for monetary policy stance, which affects capital investment and potentially the composition of federal spending through budget pressures.¹⁷ We collect both series from the Federal Reserve Bank of St. Louis (FRED) database.

¹⁶ National Center for Science and Engineering Statistics, “Definitions of Research and Development: An Annotated Compilation of Official Sources,” NCSES 22-209, May 19, 2022.

¹⁷ Federal budget aggregates such as federal debt-to-GDP, deficit-to-GDP, or total discretionary spending are not included as separate macroeconomic controls. These variables most plausibly enter R&D allocation through the overall scale of discretionary appropriations. Because our outcome variables are shares of total federal R&D, which sum to one by construction, this level channel is mechanically absorbed by the shares formulation.

Descriptive Statistics

Table 1 presents summary statistics for the main variables in our analysis over the 1956-2022 sample period.

Table 1. Summary Statistics

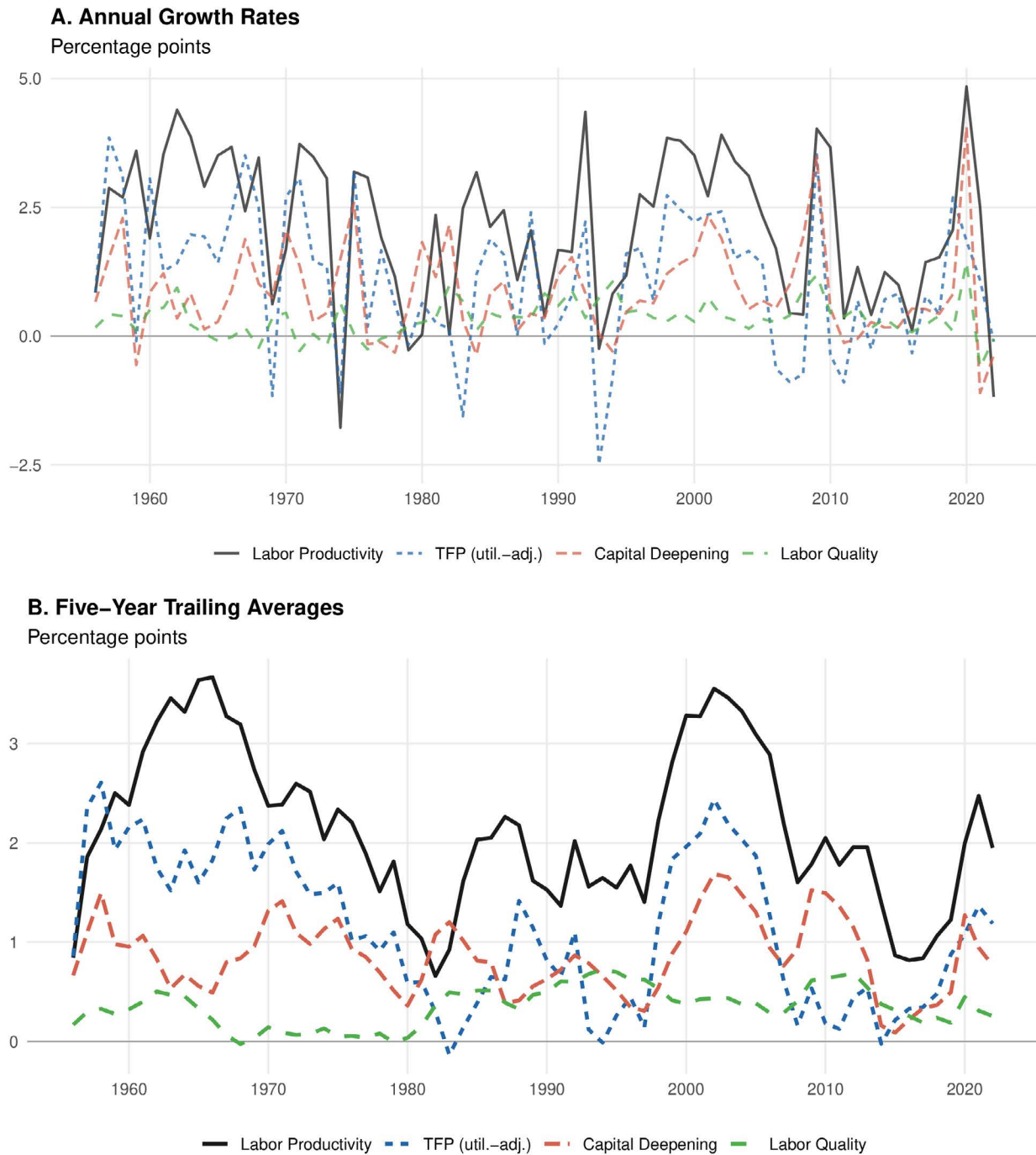
Variable	Mean	Std. Dev.	Min	Max
Labor Productivity Growth	2.13	1.45	-1.78	4.84
TFP Growth	1.11	1.39	-2.49	3.85
Capital Deepening	0.85	0.92	-1.11	4.07
Labor Quality Growth	0.34	0.36	-0.60	1.44
Basic Research Share	0.17	0.06	0.06	0.27
Applied Research Share	0.20	0.04	0.12	0.30
Development Share	0.59	0.08	0.40	0.75
Plant Share	0.04	0.02	0.02	0.10

Notes: Sample period is 1956-2022 (N=67). Productivity variables are expressed in percentage points of annual growth. Federal R&D report, and shares sum to one.

SOURCES: Authors' calculations from NSF and OMB historical tables.

The average composition of federal R&D over this period shows that development has consistently dominated, accounting for roughly 58 percent of total federal spending on average. Applied research represents about 20 percent, basic research about 17 percent, and R&D plant about 4 percent. These averages mask considerable variation over time, as illustrated in Figure 1.

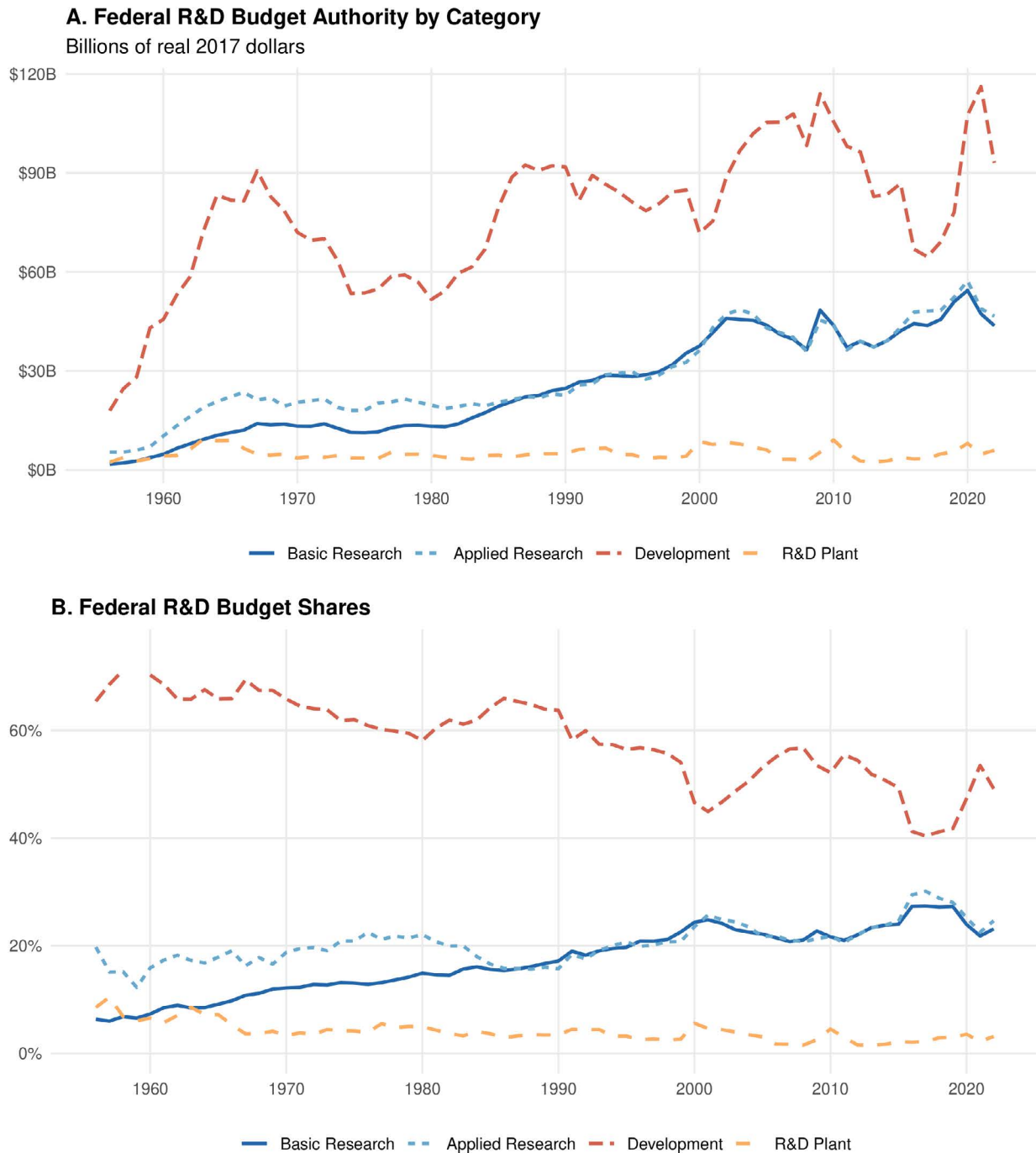
Figure 1. U.S. Productivity Growth and Components



NOTES: The four series are labor productivity growth in the U.S. business sector, utilization-adjusted TFP, capital deepening (the contribution of capital per hour growth, weighted by capital's share of income), and labor quality. All values are in percentage points of annual growth. The top panel shows the growth rates and the bottom panel shows the five-year trailing averages.

SOURCE: Fernald (2014)

Figure 2. Federal R&D Spending Composition



NOTES: Panel A reports federal R&D budget authority by activity type in billions of 2017 dollars, deflated using the GDP deflator. Panel B reports each category's share of total federal R&D.
SOURCE: NSF and OMB historical tables.

Figure 2 presents federal R&D spending in real dollars (Panel A) and as budget shares (Panel B). Development has dominated federal R&D portfolio throughout the sample period, though its

share has declined from nearly 70 percent in the early 1960s to about 50 percent by 2022. Basic and applied research shares have risen correspondingly.

Historical Variation in R&D Composition

The composition of federal R&D has evolved substantially over the study period, driven by shifting national priorities and the changing structure of the federal research enterprise. During the late 1950s and 1960s, the Space Race and Cold War competition drove substantial increases in development spending, particularly for missile systems and space vehicles (Corrado et. al. 2023). The basic research share declined during this period as development activities absorbed an increasing fraction of the federal R&D budget. The 1970s saw a partial reversal, with the development share declining as the Apollo program wound down and defense spending decreased relative to GDP. The basic research share recovered somewhat during this period. The Reagan defense buildup of the 1980s again increased the development share, though not to the levels of the 1960s.

More recent decades have seen a gradual increase in the basic research share, driven partly by the growth of health research funding through the National Institutes of Health (NIH). The doubling of the NIH budget between 1998 and 2003 represented primarily basic and applied research rather than development, shifting the overall federal composition toward earlier-stage activities.

Table 2 summarizes how the average composition of the federal R&D portfolio has shifted across broad historical eras, reflecting these successive shocks to national priorities.

Table 2. R&D Composition Has Historically Been Weighted toward Development, with Variation

Era	Basic	Applied	Development	Plant	Total R&D spend (Mean, 2017 \$B)
1956-1970	0.09	0.17	0.68	0.06	\$90.2
1971-1990	0.14	0.19	0.62	0.04	\$109.2
1991-2010	0.21	0.21	0.54	0.03	\$170.1
2011-2022	0.24	0.25	0.48	0.02	\$180.4

NOTES: This table reports time-period means of federal R&D activity-type shares and total real federal R&D budget authority across the years in the indicated era.

SOURCE: Authors' calculations from NSF and OMB historical tables.

This variation provides the empirical foundation for our analysis. The geopolitical and legislative origins of the major compositional shifts are important for identification, because changes driven by the Sputnik shock or Congressional appropriations mandates are more plausibly orthogonal to contemporaneous productivity dynamics than compositional drift driven by, for example, perceived returns to particular research investments.

That said, there are two relevant concerns for interpretability. First, multi-year legislated funding buildups may be partially foreseeable before they appear in realized budget authority. This dynamic raises the possibility that measured composition changes partly reflect anticipated, rather than unforeseen, policy shifts. Due to this concern, we vary the lag lengths in our robustness analysis, finding broadly similar results. Second, the major compositional episodes coincide with distinctive macroeconomic eras (e.g., the defense-heavy 1960s overlap with the postwar productivity boom, and the NIH buildup with the late-1990s technology expansion), which could confound estimated associations if those eras differ along dimensions unaccounted for in our specifications. We address this concern by allowing for trends to vary by policy regime, again finding broadly similar results.

A Local Projections Framework to Isolate R&D Spending and Compositional Shocks

This section describes our approach to estimating the dynamic relationships between federal R&D composition and productivity outcomes. Building on Jordà (2005), we use a Local Projections framework to identify shocks in R&D spending. We construct R&D composition shocks, which we interpret as directionally informative.¹⁸ This section further details our approach.

Local Projections Framework on Residualized R&D Spending Shocks

We estimate dynamic responses using the Local Projections method introduced by Jordà (2005). For each horizon hh from 00 to HH years, we estimate a separate regression of the form:

$$y_{t+h} = \alpha_h + \beta_h \cdot z_t + \gamma_h \gamma_h'^{X_t} + \varepsilon_{t+h},$$

where y_{t+h} denotes a productivity outcome hh years after the shock, z_t is the standardized R&D composition innovation, X_t is a vector of controls including lags of the dependent variable, the output gap, total R&D spending, and the change in the federal funds rate, and ε_{t+h} is an error term.

The coefficient β_h represents the response of the productivity outcome at horizon hh to a one-standard-deviation shock in R&D composition. By estimating separate regressions for each horizon rather than specifying a complete dynamic system, Local Projections avoid potential misspecification of the transmission mechanism. If the true dynamics involve complex or nonlinear propagation, Local Projections will capture the reduced-form response at each horizon without requiring the correct specification of the intermediate steps.

We compute cumulative responses as the sum of horizon-specific coefficients:

¹⁸ Estimates are bounded using the moving block bootstrap procedure.

$$\text{Cumulative}_H = \sum_{h=0}^H \beta_h.$$

This cumulative measure captures the total effect on the level of productivity over the horizon, aggregating the period-by-period contributions into a single summary statistic. For our main results, we focus on the ten-year cumulative response, which allows time for the effects of R&D composition changes to materialize while maintaining reasonable sample sizes for estimation. We evaluate alternative horizons as robustness tests.

Our analysis requires variation in R&D composition that represent unpredictable changes from the perspective of economic agents. We construct these shocks as residuals from forecasting equations, in the spirit of Ramey (2016).¹⁹

For each R&D composition variable, we estimate a forecasting equation that includes four lags of the composition variable itself, four lags of log total R&D, the output gap, and the change in the federal funds rate (FFR). For share variables, we additionally include contemporaneous log total R&D and the residual from the total R&D forecasting equation. The shock is defined as the residual from this forecasting regression, representing the component of compositional change that could not have been predicted based on available information.²⁰

Formally, for a share variable $s_{j,t}$ representing the share of R&D type jj , we estimate:

$$s_{j,t} = \phi_0 + \sum_{k=1}^4 \phi_k s_{j,t-k} + \sum_{k=0}^4 \psi_k \ln(\text{Total}_{t-k}) + \theta_1 \text{gap}_t + \theta_2 \Delta \text{FFR}_t + \theta_3 v_t^{\text{total}} + u_{j,t},$$

where v_t^{total} is the shock to total R&D constructed analogously. The residual $u_{j,t}$ serves as our R&D compositional shock, which we standardize to have mean zero and unit variance for comparability across specifications.

The inclusion of the total R&D shock in the share equations orthogonalizes compositional shocks with respect to aggregate R&D movements. This ensures that our share shock constructions capture pure reallocation effects rather than changes in composition that mechanically accompany aggregate R&D expansion or contraction.

¹⁹ Unlike the “news” measures emphasized in Ramey (2016), which typically involve news about future government spending, our shocks are not direct measures of forward-looking policy announcements or expectations. Accordingly, we interpret them as unexpected realized compositional changes rather than pure news about future policy paths. We emphasize that unpredictability conditional on observable factors does not rule out correlation with policymakers’ private information or other omitted variables.

²⁰ Our R&D composition innovations are constructed as residuals from forecasting regressions. Because these forecasting equations are estimated on the full 1956-2022 sample, the resulting residuals are in-sample deviations from an ex-post best fitting predictive model, rather than a literal one step ahead forecast error formed only with information available at time $t - 1$. A stricter implementation would recursively reestimate the forecasting equations on an expanding window ending at $t - 1$ and define the shock in year t as the resulting forecast error, which removes any look-ahead in coefficient estimation. We do not implement this recursive construction here because the annual sample contains only 67 observations so early sample shocks would be based on very short estimation windows and would be comparatively noisy once standardized, materially reducing the power in subsequent local projection estimates.

Substantively, the residual variation that constitutes our composition shocks plausibly reflects unanticipated changes in federal R&D priorities. These include shifts in administration and congressional priorities, programmatic reorientation within agencies such as new initiatives or sunset programs, crisis-driven reallocations in response to events such as recessions or public health emergencies, and budget negotiation outcomes that diverge from the trajectory predicted by recent history.

Figure A1 in the appendix presents the empirical distribution of each standardized composition shock series alongside a standard normal density for reference. The distributions are reasonably symmetric, centered near zero, and broadly consistent with normality, which supports our use of normal-based inference for cumulative responses.

Identifying Assumptions and Limitations

A causal interpretation of our local projection estimates rests on a conditional exogeneity assumption that the constructed R&D composition shocks are uncorrelated with unobserved determinants of future productivity, conditional on the information set included in our forecasting equations and outcome regressions.

We operationalize R&D composition shocks as residuals from forecasting regressions for each composition series conditioned on prior investments, lagged total R&D spending, and other macro controls. These controls ought to account for high-level macroeconomic trends, such as the Federal Funds rate, that might influence federal spending decisions. By construction, these shocks are thus orthogonal to the included predictors.

The identification claim is that, to the extent this conditioning is a sufficient proxy for the information policymakers use when adjusting the R&D portfolio in response to unexpected macroeconomic and productivity-relevant conditions, the remaining residual variation can be interpreted as an as-if random policy surprise, or “shock” (Ramey 2016).

Several features of our design strengthen this interpretation. First, federal budgeting and program execution operate with substantial institutional lags.²¹ Compositional shifts in realized budget authority typically reflect plans and negotiations initiated in earlier years and are implemented over multiple years. This timing reduces the plausibility that our composition shocks are mechanically responding to contemporaneous realized productivity shocks. Second, our preferred productivity measure is utilization-adjusted TFP, which mitigates a prominent source of spurious correlation of cyclical fluctuations in measured productivity that could be jointly related to fiscal conditions. Third, for share shocks we explicitly orthogonalize compositional shocks with respect to aggregate R&D movements by including contemporaneous

²¹ The budget process has been delayed since 2003 and half the time has resulted in one or more continuing resolutions or government shutdowns (CRFB, “The Budget Act at 50 – Budget Process Reform Recommendations for the 118th Congress,” October 17, 2023; Congress.gov, “The Congressional Budget Process Timeline.” Library of Congress, 20 February 2026, <https://www.congress.gov/crs-product/R47235>).

total R&D and the total R&D shock in the share forecasting equations. This step pushes identification toward within-portfolio reallocation rather than changes in the overall fiscal stance, helping separate compositional shifts from broad expansions or contractions that may be more tightly linked to macroeconomic conditions.

However, these arguments do not eliminate threats to identification. The most important limitation is that unpredictability relative to our conditioning set does not guarantee exogeneity. Composition may still move in response to anticipated future productivity or to policymakers' private information about future conditions that is not captured by our observed controls. More broadly, unobserved policy changes or geopolitical and mission-priority shocks (e.g., shifts in defense posture, regulatory regimes, or procurement strategy) could affect both the federal R&D mix and productivity through channels outside measured R&D composition, creating potential confounding. An omitted shock would need to occur at the same realization date as the constructed R&D shock, be unpredictable from the lagged variables in our forecasting model and operate on productivity through channels other than the federal R&D portfolio it influences.

For greater clarity on how possible omitted shocks shape the interpretation of our results, consider two potential examples: geopolitical shifts and R&D tax expenditures. First, unanticipated geopolitical events can move federal R&D priorities and aggregate productivity in the same year through channels that operate alongside the R&D portfolio itself. The relevant failure mode is an event that arrives as a surprise to the forecasting model, prompts a contemporaneous reallocation of federal R&D, and affects productivity through trade, supply chains, regulation, capital investment, or business sentiment. A geopolitical shift telegraphed in advance would typically be absorbed by the lagged predictors and would not necessarily bias our estimates. Second, our R&D spending measures capture federal outlays and excludes R&D tax expenditures. The federal R&D tax credit acts as an alternative form of public support for R&D, and discrete changes to the credit alter the public subsidy environment in ways that our outlays variable does not capture. Bias from this source arises only when changes to tax expenditures are correlated with our outlay shocks at the same realization date and have an independent productivity effect through the private R&D they stimulate. The sign of any such bias follows the sign of that co-movement, which we cannot determine within our current specifications. For these reasons, we interpret our findings as informative associations about the effect of public outlays on R&D driven productivity that warrant further investigation. We probe the plausibility of the conditional exogeneity assumptions in several ways. First, we document that the shocks have the statistical properties expected of forecast errors, including minimal serial correlation and limited predictability from expanded information sets. Second, we conduct lead/anticipation tests to assess whether future composition shocks predict earlier productivity movements, as significant lead relationships would be consistent with anticipation, reverse causality, or omitted variables. Third, we evaluate the robustness of the results to alternative lag lengths, specifications, and subsamples to assess whether the results hinge on narrow episodes or

modeling choices. These diagnostics, discussed below, cannot prove exogeneity, but they help bound the most direct failures of the identifying assumptions.

Overall, our preferred interpretation of our results is that under a sufficient information set assumption (i.e., that our forecasting controls approximate the information set that systematically links R&D portfolio choices to expected productivity-relevant conditions) our shocks can be interpreted as unexpected reallocations in the federal R&D mix that are plausibly exogenous with respect to subsequent productivity shocks. Where this assumption fails, our estimates should be read as reduced-form dynamic associations rather than definitive causal effects.

Bounding the Estimates: Identifying Standard Errors for Horizon Coefficients and Cumulative Effects

We employ two approaches to estimating and bounding coefficients, appropriate for different quantities of interest. For horizon-specific coefficients, we use Newey-West heteroskedasticity and autocorrelation consistent standard errors with automatic bandwidth selection.²² The bandwidth is computed as:

$$L = \lfloor 4 \times (n/100)^{2/9} \rfloor$$

where n is the sample size for the regression at horizon h . This bandwidth increases appropriately with horizon to account for the additional serial correlation induced by overlapping observation windows.

For cumulative responses, which sum the coefficients across horizons, we employ the moving block bootstrap procedure recommended by Jentsch and Lunsford (2019).²³ Standard asymptotic inference for cumulative impulse responses can be unreliable because the cumulated sum of coefficients has a nonstandard distribution when the underlying series are persistent. The moving block bootstrap preserves the dependence structure of the data by resampling contiguous blocks of observations.

Our implementation draws 600 bootstrap replications with block length 5, representing approximately 7-8 percent of the sample. For each replication, we reconstruct all R&D composition shocks from the resampled data and re-estimate the full set of Local Projections. The bootstrap standard error for the cumulative response is computed as the standard deviation across replications, and confidence intervals are constructed using the normal approximation with the bootstrap standard error.

²² Whitney K. Newey and Kenneth D. West, "A Simple, Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix," *Econometrica*, Vol. 55, No. 3, pages 703-708, May 1987.

²³ Also, see Efstathios Paparoditis, "Bootstrapping autoregressive and moving average parameter estimates of infinite order vector autoregressive processes," *Journal of Multivariate Analysis*, Vol. 57, No. 2, p. 277-296, 1996; Óscar Jordà & María Dolores Gadea, "Local Projections Bootstrap Inference," Federal Reserve Bank of San Francisco Working Paper 2025-21, September 2025.

Basic Research Shows the Largest Productivity Gains; Development Is Associated with Declines

This section presents our main empirical findings. While overall research spending is not significantly associated with productivity due to large standard errors, we do find significant compositional effects. In particular, basic research is associated with the largest gains in total factor productivity and labor productivity, while development is associated with declines. These relationships directionally hold for log level changes to individual categories. Simulating bilateral composition shifts from one category to another, we find that expected productivity increases when funding is reallocated from development to any other research type, with the greatest point estimates when reallocated toward basic research. In this section, we describe our results in more detail.

Aggregate R&D Effects

Before examining composition, we estimate the productivity response to shocks in total federal R&D regardless of its allocation across categories. Table 3 reports cumulative impulse responses over a ten-year horizon, which is the typical Congressional Budget Office (CBO) budget window (Budget Committee, 2024), for each productivity component.

Table 3. Total R&D Shock Effects (Cumulative at h=10)

Outcome	Estimate	Bootstrap SE	95% CI
TFP	0.88	0.67	[-0.42, 2.19]
Capital Deepening	0.16	0.40	[-0.63, 0.95]
Labor Quality	0.07	0.19	[-0.31, 0.45]
Labor Productivity (Total)	1.11	0.88	[-0.62, 2.84]

Notes: Estimates are cumulative percentage point responses over ten years to a one-standard-deviation innovation in total federal R&D. Bootstrap standard errors from 600 moving block bootstrap replications with block length 5.

The estimated cumulative labor productivity response to an aggregate R&D shock is 1.1 percentage points, with approximately two-thirds of this effect attributed to TFP and the remainder split between capital deepening and labor quality. However, the bootstrap standard error of 0.88 percentage points implies that this estimate is not statistically distinguishable from zero at conventional significance levels.

This imprecision in aggregate responses could potentially be explained by compositional differences. If different types of R&D have heterogeneous productivity effects, aggregate shocks that combine these types will reflect a weighted average that may be difficult to detect precisely.

Moreover, the variation in aggregate R&D may be driven by factors that simultaneously affect productivity through other channels, confounding the R&D effect.

R&D Share Shock Effects

Compositional variation, by contrast, may provide cleaner identification to the extent that the allocation of a given R&D budget reflects different considerations than its overall level, conditional on prior allocations and broad macroeconomic trends. We turn now to share shocks, which capture the effects of reallocating R&D across categories—for example, from basic research to development research, or vice versa. The overall level of R&D spending is tightly linked to fiscal conditions, macroeconomic cycles, and broader political priorities, all of which plausibly affect productivity through multiple channels simultaneously. The allocation of a given R&D budget across categories, however, may reflect a more specific set of decisions involving agency-level program choices, congressional earmarks and appropriations subcommittee preferences, the maturation of particular research agendas, and shifts in mission emphasis within agencies. Based on this understanding, these allocation decisions would be less obviously correlated with aggregate macroeconomic conditions that independently drive productivity.

Our primary specifications examine shocks to the shares of each R&D type in the federal portfolio. These share shocks capture the effect of reallocating R&D toward a particular category while implicitly reducing allocations to other categories, holding total R&D constant. Table 4 reports cumulative ten-year responses.

Table 4. R&D Share Shock Effects (Cumulative at year 10)

Research category	TFP	Capital	Labor Quality	Labor Productivity
Basic Share	1.51** (0.73)	0.60 (0.40)	-0.02 (0.19)	2.09** (0.93)
Applied Share	1.10 (0.72)	0.86** (0.40)	-0.25 (0.19)	1.70* (0.92)
Development Share	-1.36* (0.73)	-0.92** (0.40)	0.04 (0.19)	-2.23** (0.93)
Plant Share	0.96 (0.70)	0.77* (0.40)	-0.35* (0.20)	1.38 (0.87)

Notes: Estimates are cumulative percentage point responses over ten years to a one-standard-deviation shock to the share innovation, defined as the residual from the share forecasting equation. Note that the coefficients in the TFP, capital deepening, and labor quality column sum to the value in the Labor Productivity column (with small deviations due to rounding). Bootstrap standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The estimates reveal substantial heterogeneity across R&D types. Shocks that increase the basic research share are associated with the largest positive effects on labor productivity, with a cumulative response of 2.09 percentage points, significant at the five percent level.

Approximately three-quarters of this effect operates through total factor productivity, suggesting that basic research primarily enhances the efficiency with which the economy combines inputs rather than stimulating capital accumulation directly. Basic research itself spans a range of activities, from fundamental theoretical work to mission-oriented investigation. Our aggregate

measure cannot distinguish among these subcomponents, which may have distinct productivity returns.

Shocks that increase the applied research share also show positive associations with labor productivity, with a cumulative response of 1.70 percentage points, though this is only significant at the ten percent level. The composition of this effect differs somewhat from basic research, with a relatively larger contribution from capital deepening. This pattern is consistent with applied research generating more immediate and directed effects on investment opportunities than the diffuse knowledge gains from basic research.

Shocks that increase the development share are associated with cumulative labor productivity declines of 2.23 percentage points over ten years. Estimated channels for this decline include both TFP and capital deepening, as opposed to labor quality, where the change is not statistically significant. The negative association implies that reallocating federal R&D toward development is associated with productivity losses roughly comparable in magnitude to the gains from reallocating toward basic research. Because shares sum to one, the negative development coefficient mechanically includes the foregone return from categories with higher estimated returns; we cannot separately identify the direct negative effect attributable to development spending itself. Interpreted within the share framework, this finding implies that the current federal allocation places more weight on development than would maximize labor productivity growth. Given this constraint, we interpret this finding as providing a rank-ordering of associations with productivity, with development spending as the least contributive toward labor productivity outcomes. This finding is consistent with prior literature, which generally finds basic research to contribute more to productivity relative to applied or development research (as reviewed in Hall et. al. 2009). Potential reasons for lower returns to development spending that would be consistent with prior literature include an emphasis on defense applications (Fieldhouse & Mertens 2024, 2025), reduced spillovers associated with greater IP protections (Dyevre 2024), and lower creative flexibility and tolerance for failure relative to basic research conditions (Azoulay et. al. 2009).

Shocks to the R&D plant share (i.e., the share of R&D spending for research facilities and equipment) show positive labor productivity responses, though they are less precisely estimated than responses of other shares. Capital deepening is the only positive channel that is statistically significant for the R&D plant share. The TFP channel has a comparable point estimate but is imprecisely estimated. This pattern is consistent with research infrastructure investments directly augmenting the capital stock available for productive use, in addition to any indirect effects through enhanced research capabilities, net of depreciation.²⁴

²⁴ Federal R&D plant aggregates a heterogenous set of investments. Capital intensity likely varies substantially across scientific fields, with much higher costs in disciplines such as physics, space science, or large biomedical infrastructure than they are in more software-heavy fields. Our coefficient on the plant share is a portfolio weighted average of the underlying disciplinary effects, with weights determined by historical allocation. The data report

The approximate symmetry between the basic research and development share responses partly reflects the simplex constraint. Because the shares sum to one, a positive shock to the development share implies offsetting decreases in other categories. As such, we cannot conclude that development spending is actively harmful, only that it appears to have the lowest returns among the four categories. Qualitatively, the pattern implies a rank ordering of net labor productivity returns, with basic research at the top and development at the bottom among the four categories. The bilateral reallocation specifications (Table 6) provide a more direct test of this ordering and produce consistent results.

Effects of Shocks to R&D Funding Levels

An alternative specification examines shocks to the log level of each R&D type, controlling for the levels of other types. This approach captures the effect of increasing one type of R&D spending while holding other types constant, rather than the reallocation implicit in share shocks. Table 5 reports results.

Table 5. R&D Level Shock Effects (Cumulative at year 10)

Innovation	TFP	Capital	Labor Quality	Labor Productivity
log Basic	1.26* (0.66)	0.17 (0.39)	-0.19 (0.21)	1.24 (0.85)
log Applied	1.44** (0.72)	0.74* (0.44)	-0.09 (0.20)	2.09** (0.95)
log Development	0.01 (0.81)	-0.65 (0.49)	0.02 (0.22)	-0.63 (1.09)
log Plant	0.97 (0.71)	0.88** (0.40)	0.02 (0.19)	1.87** (0.88)

Notes: Estimates are cumulative percentage point responses over ten years to a one-standard-deviation shock in the log level. Bootstrap standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The level specifications confirm the qualitative patterns from the share analysis while showing somewhat smaller magnitudes. The attenuated effects likely reflect that level shocks do not embed the implicit reallocation from other categories that share shocks capture. When basic research increases while other categories remain constant, the overall R&D portfolio becomes more basic-intensive; when basic research increases as part of a general expansion that maintains constant shares, the compositional effect is muted.

The sign pattern across R&D types remains consistent. Basic and applied research show positive associations with labor productivity, development shows negative associations, and plant investments show positive effects. The relative magnitudes differ across specifications. In the level specification, applied research has the largest cumulative response and basic research drops to third. The larger applied research coefficient in the level specification is consistent with

federal R&D by activity type, and we do not separately identify discipline-specific effects within the plant category as this would require more fine-grained data.

applied research being particularly productive when paired with parallel investment in other R&D categories.

Direct Reallocation Effects Reveal Opportunity for Productivity Gains

The share and level specifications examine each R&D category individually. To provide direct estimates of the productivity consequences of bilateral reallocation, we construct additive log-ratio shocks that measure the effect of shifting R&D from development toward other categories. These specifications isolate bilateral reallocation effects. Table 6 reports results.

Table 6. Reallocation Shock Effects: Productivity Gains from Shifting Away from Development (Cumulative at h=10)

Reallocation	TFP	Capital	Labor Quality	Labor Productivity
Basic vs. Development	1.64** (0.75)	0.84** (0.41)	-0.07 (0.19)	2.42** (0.95)
Applied vs. Development	1.36* (0.73)	0.95** (0.40)	-0.19 (0.19)	2.12** (0.92)
Plant vs. Development	0.77 (0.69)	1.08*** (0.40)	-0.02 (0.19)	1.83** (0.86)

Notes: Estimates are cumulative percentage point responses over ten years to shocks that reallocate R&D from development toward the indicated category. Bootstrap standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The reallocation estimates confirm that shifting federal R&D away from development toward any other category is associated with substantial productivity gains. Reallocation toward basic research shows the largest response for labor productivity at 2.42 percentage points over ten years, followed by reallocation toward applied research at 2.12 percentage points and reallocation toward plant at 1.83 percentage points. These estimates are economically meaningful and statistically significant. To put this in context, the annual average labor productivity growth over our sample period is approximately 1.8 percent, so a reallocation shock that generates 2.4 additional percentage points over ten years represents a cumulative gain comparable to more than one year of normal productivity growth.

Core Results Hold Across Lag Structures, Subsamples, and Placebo Tests

This section assesses whether the compositional estimates in Section 5 reflect unexpected reallocations in the federal R&D portfolio rather than (1) predictable dynamics in the composition series, (2) anticipation or reverse causality (pre-trends), or (3) sensitivity to a small set of episodes or modeling choices. Because our shocks are constructed as forecast residuals, these exercises cannot prove exogeneity. They can probe the most direct ways in which the identifying assumption could fail.

We focus the main text on the most consequential checks and summarize them in a single table (Table 7). Appendix A reports the full robustness results.²⁵

Entries in columns 1 through 3 are the cumulative ten-year labor productivity response to a one-standard-deviation shock to the indicated R&D share, estimated using the same LP specification as in Section 5. Column 4 reports p-values from lead (anticipation) tests, and column 5 reports p-values from a circular-shift placebo exercise that preserves the shock sequence but breaks its historical timing.

Table 7. Robustness Summary

Shock (share)	Baseline (4 lags)	2 lags	6 lags	Lead test p-value (h=10)	Placebo p-value
Basic share	2.09**	2.15**	1.10	0.083	0.036
Applied share	1.70*	1.68*	1.05	0.250	0.023
Development share	-2.23**	-2.26**	-1.56	0.340	0.028
Plant share	1.38	1.37*	2.01**	0.080	0.304

NOTES: Baseline point estimates correspond to the share-specific results reported in Table 3. Alternative lag estimates correspond to re-estimation under the indicated lag length. The fourth column reports the p-value from a joint test that leads of the shock have no predictive power for the outcome through horizon $h=10$; failure to reject (high p-values) is consistent with the shock being unanticipated. The last column is from a randomization test that reassigns shock timing across years and reports the share of placebo replications yielding a cumulative response at least as large in absolute value as the baseline estimate; small values indicate that the baseline effect is unlikely under the null of no effect. (Appendix A reports full results and associated inference). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Our baseline specification uses four lags in both shock construction and LP estimation as a compromise between capturing medium-run persistence and preserving degrees of freedom in a 67-year annual sample. Table 7 shows that the ten-year cumulative labor productivity responses are qualitatively similar under shorter lag structures (2 lags), but under long lag structures (6 lags) the estimates generally attenuate. The attenuation is consistent with longer lag specifics absorbing more low frequency variation.

A central threat in policy time series designs is that shocks may proxy for slow-moving changes or anticipation effects. Productivity might begin moving before the measured shock, or policy changes may respond to information about future productivity not captured in our controls. We therefore implement lead tests that include three-year leads of each shock in the LP regressions and test whether those leads are jointly significant.

Column 5 of Table 7 reports p-values for the joint lead test at the ten-year horizon. In general, we find no strong evidence that future composition shocks systematically predict earlier productivity movements. The plant share and basic share estimates are marginal cases,

²⁵ A minimal requirement for interpreting our forecast residuals as innovation reallocations is that they do not exhibit strong remaining predictability. Appendix A reports serial-correlation diagnostics for each innovation. Across all composition innovations (and total R&D) we fail to reject the null of no serial correlation at conventional levels, consistent with the shocks behaving like forecast errors rather than slow-moving trends.

suggesting that plant and basic reallocations could be more gradual, more correlated with slow moving drivers, or measured with different timing than the other components.

Finally, we ask whether the estimated cumulative responses are tied to the historical timing of composition reallocations or would arise mechanically from generic persistence. We implement a placebo test based on circularly shifting the realized shock series by a random offset in calendar time, re-estimating the same LP specification each time. This placebo preserves the realized shock sequences (including its serial correlation and low frequency variation) while breaking its alignment with historical productivity movements.

Column 6 of Table 7 reports placebo p-values for labor productivity. The placebo test re-estimates each shock's effect after shifting its timing to other (non-actual) years. The p-value reports how unusual the actual-timing estimate is relative to the distribution of placebo estimates. A small p-value indicates that the actual-timing estimate lies in the tails of the placebo distribution, which means the result is tied to the specific historical timing of R&D reallocations and is unlikely to reflect generic persistence or comovement patterns. The basic, applied, and development share estimates are unusually large relative to the distribution under shifted timing, with p-values around 0.02 to 0.04, consistent with these results being meaningfully tied to the timing of actual reallocations. In contrast, the plant share estimates are not unusually large under this placebo, again suggesting that the plant results are the least timing-specific and should be interpreted with greater caution. Appendix A reports the full placebo results for TFP, capital deepening, and labor quality.

Stability Check Across Structural Break: Bayh-Dole Act of 1980

To examine whether our estimated associations are stable over time, we re-estimate the models allowing for the shock coefficient to vary before and after 1980. We focus on 1980 as the break year due to the Bayh-Dole Act of 1980, which fundamentally restructured incentives for U.S. R&D. Prior to the Act, government agencies retained patent rights to federally funded research. The Bayh-Dole Act allowed universities and publicly funded researchers to patent and commercialize their inventions, granting private industry access to licensing agreements for exclusive use. This shift catalyzed academic-industrial collaboration and accelerated technology transfer (Loise and Stevens 2010; CSIS 2022).

Robustness results are reported in Appendix A. Overall, the estimates appear stable across this break, with 17 of 20 estimates not differing significantly between periods at the five percent level. The three exceptions occur at the zero horizon, where standard errors are relatively narrow. It is possible that this shift reflects a change in the relationship, though estimates appear to converge over time. Basic research prior to 1980 does appear to contribute more to labor productivity and innovation by year ten, but the difference is not statistically significant. One possibility is that this difference reflects the mission-oriented funding of the 1960s and 1970s and complementarities with the substantial baseline funding level and structural support for public R&D. Our robustness check is consistent with findings of a general decline in R&D

effectiveness over time, which appears specifically in how basic research spending contributes to total factor productivity and labor productivity in the longer-term (Corrado et. al. 2023). At the same time, large standard errors and time horizons make it difficult to distinguish between genuine structural change and estimation uncertainty.

A Portfolio Framework Suggests Public Investment in Development at the Cost of Productivity

Building on the estimated productivity associations documented in the preceding sections, we now develop a portfolio optimization framework to evaluate alternative allocations of federal R&D. We emphasize that this exercise requires strong assumptions about the stability and causal interpretation of our estimates, and we present it primarily as an illustrative tool for thinking about compositional tradeoffs rather than a definitive policy recommendation. The exercise also requires a specific focus on federal R&D as a productivity-enhancing investment, which ignores potential other policy-relevant objectives of R&D spending.

Framework

We conceptualize the allocation of federal R&D across categories as a portfolio choice problem analogous to asset allocation in finance. Let $w = (w_B, w_A, w_D, w_P)$ denote the shares of R&D allocated to basic research, applied research, development, and plant, respectively, with shares constrained to be non-negative and sum to one.

The expected productivity return from portfolio w is computed as:

$$\mu(w) = w_B\mu_B + w_A\mu_A + w_D\mu_D + w_P\mu_P,$$

where μ_j denotes the cumulative productivity effect estimated for R&D type jj . To aggregate across productivity components, we construct a composite index:

$$F(w) = 0.5 \times \mu^{TFP}(w) + 0.3 \times \mu^K(w) + 0.2 \times \mu^{LQ}(w),$$

weighting total factor productivity (weighted at 0.5), capital deepening (weighted at 0.3), and labor quality (weighted at 0.2), with weights selected to roughly reflect each component's average contribution to labor productivity growth over our sample period (authors' calculations from Fernald 2014). Our robustness analysis treats these proportions as a benchmark and evaluates portfolio performance across a Dirichlet distribution of weights centered on the baseline as well as a discrete set of alternative weighting schemes, spanning equal weights, single-component weights, and growth-accounting weights.

The uncertainty in estimated effects generates risk that portfolios must manage. We characterize each portfolio by three summary statistics: (1) the mean expected return; (2) the downside expected return, taken as the 5th percentile of the portfolio's return distribution; and (3) the probability that the portfolio generates a positive return. Each summary reflects both the bootstrap standard errors of the asset-level impulse responses and the choice of weights used to aggregate productivity components into a scalar score. Portfolios that score highly on the

downside expected return criterion remain attractive in two senses. First, by construction, they continue to generate positive productivity returns even when the underlying asset-level impulse responses are realized below their point estimates in the bootstrap distribution. Second, they continue to generate positive productivity returns when the weights used to aggregate productivity components into a composite score are varied within the bounds. The downside expected return therefore serves as a single summary measure of portfolio performance that is robust to both sources of uncertainty.

Sampling and Evaluation

We draw 10,000 random portfolios uniformly from the four-dimensional simplex and evaluate each portfolio's expected return, standard deviation, and downside risk. For each portfolio, we conduct 1,000 Monte Carlo iterations. In every iteration, we redraw the asset-level cumulative impulse responses from their bootstrap sampling distributions, and we redraw the outcome weights from a Dirichlet distribution centered on the baseline proportions. The resulting 1,000 returns per portfolio yield the three summary statistics described above. We identify the Pareto optimal portfolios that are not dominated by any other portfolio in the three-dimensional space of mean returns to TFP, capital deepening, and labor quality. We also identify the top decile of portfolios ranked by their downside expected return.

About 8 in 10 Randomly Drawn Portfolios Outperform the Status Quo

Table 8 summarizes the distribution of portfolio performance across different selections.

Table 8. Portfolio Performance by Selection Criterion Show Room to Increase Productivity

Portfolio Set	N	Mean Return [5 th , 95 th percentile]	Mean Downside	Pr(Return > 0)
All Random	10,000	0.35 [-0.07, 0.79]	-0.07	0.80
Pareto Frontier	70	0.54 [0.01, 1.08]	0.01	0.81
Top 10% by Downside	1,000	0.77 [0.32, 1.25]	0.32	1.00

Notes: Return is the composite productivity index, $F(w)$. Downside is the 10th percentile of the bootstrap distribution. Pr(Return > 0) is the fraction of bootstrap replications yielding positive returns.

Approximately 80 percent of random portfolios produce positive expected returns at the Monte Carlo mean. The average random portfolio produces a 0.35 percentage point cumulative ten-year productivity gain relative to the baseline portfolio, though the 90-percent confidence interval spans zero. The Pareto optimal portfolios identified from the Monte Carlo improve on the sample mean and produce a cumulative score of 0.54 percentage points and a 90 percent confidence interval between 0.01 and 1.08 percentage points. Portfolios in the top decile produce a mean return of 0.77 percentage points, with a 90-percent confidence interval from 0.32 to 1.25 percentage points. Table 9 characterizes the composition of different portfolio selections.

Table 9. Average Portfolio Composition by Selection Uniformly Shifts Away from Development

Selection	Basic	Applied	Development	Plant
Historical Average (actual)	17%	20%	59%	4%
Pareto Frontier	47% [6, 93]	31% [0, 84]	18% [0, 83]	3% [0, 10]
Top 10% by Downside	48% [23, 76]	27% [4, 54]	3% [0, 7]	22% [2, 48]

Notes: Values are average portfolio weights across portfolios in each selection. Values in the brackets are the 95 percent confidence interval around the estimate.

The composition analysis reveals differences between the historical allocation and those that maximize productivity according to our estimated productivity effects. The historical average portfolio of public R&D spending allocates 59 percent to development, while Pareto-optimal portfolios allocate only 18 percent. Conversely, the historical basic research share of 17 percent is nearly three times less than the Pareto optimal share of 47 percent. Bracketed values capture the joint effect of asset standard errors and outcome-weighted uncertainty on the bounds of each share. The robust portfolios place roughly three times the historical weight on basic research, more than five times the historical weight on plant, and almost no weight on development, with the bounds on development holding at no more than seven percent even at the 95th percentile of the joint Monte Carlo. While there are wide confidence bands, these point estimates are consistent with an overweighting of R&D spending toward development, suggesting productivity improvements to be gained from a shift toward basic R&D, in particular.

Caveats

There are several important limitations to this application of portfolio analysis. First, the optimal portfolios depend on the point estimate coefficients on productivity effects, though we consider uncertainty in the estimates using bootstrapped standard errors. To the extent that the true productivity effects differ from our estimates, optimal allocations would shift. Second, the analysis focuses on marginal effects of shifting the marginal dollar from one research type into another, assuming that effects are stable across different allocation regimes. General equilibrium considerations suggest that large reallocations could change the returns to each category, or that the true effects could be nonlinear. Third, federal R&D composition reflects objectives beyond productivity maximization, including national security, public health, and scientific capability. Development spending in particular is heavily concentrated in defense and other mission-oriented programs whose social value extends beyond measured labor productivity. Reallocating funding away from development involves tradeoffs that our productivity-focused analysis does not characterize. Compositional policy must account for both the productivity consequences we estimate and the consequences for other policy objectives.

We present the portfolio analysis primarily as a tool for illustrating the implications of heterogeneous productivity effects on the opportunity cost of high R&D funding allocation to development. The estimated optimal portfolio is not intended as a policy prescription, but as a directional indicator. The analysis clarifies that if our estimates are approximately correct, and assuming stable linear marginal effects, the productivity consequences of compositional choices are substantial enough to warrant serious attention in science and technology policy deliberations.

Our Results Support a Rebalancing of U.S. R&D Budget Priorities

Our findings support recent calls for a change to budget scoring of R&D spending, in order to better capture the different expected returns across R&D types. At the policy level, a productivity-focused Federal R&D portfolio would rely less on development spending and more on basic research; this finding ought to be considered as a tradeoff to the current development-heavy R&D budget. In this section, we elaborate on this interpretation and mechanisms.

Federal R&D Investment in Basic Research is Low, Relative to Expected Productivity Gains

Our finding of a positive association between basic research and productivity is consistent with both prior literature and theoretical expectations, including the dominant mechanism of total factor productivity. Basic research generates fundamental knowledge that can be applied across many contexts, creating broad spillovers that raise productivity throughout the economy (Nelson 1959; Dyevre 2024). Further, funding for basic research has been found to contribute to and maintain high-caliber human capital in the United States (Tham et. al. 2024). Long lags typically associated with basic research effects are consistent with our finding that basic research works primarily through TFP rather than capital accumulation, as knowledge diffusion takes time. Yet our analysis supports the view that gains at least begin to materialize within the budget window of ten years, rather than taking 20 or more years to materialize as assumed in current budget scoring approaches (Gullo et. al. 2025). The cumulative ten-year effect of approximately 2.1 percentage points per standard-deviation shock is on the higher end of prior estimates and implies substantial returns to basic research investment. This estimate, combined with our portfolio analysis, suggests significant underinvestment in basic research relative to a productivity-maximizing benchmark.

The implication of this finding for policymakers is twofold. One, standard budget scoring approaches may underweight the productivity value of basic research investment, contributing to a systematic underinvestment in this category of R&D spending relative to other categories. Updating budget scoring to reflect longer-term futures, as well as greater returns within the budget window, could help policymakers more accurately weigh the estimated benefits against the costs of basic research. Two, federal spending on basic research is a tested and effective

channel to increase economic productivity through total factor productivity. As the United States faces high and increasing budget deficits, taking an investment approach to the Federal budget would suggest a particularly high return to basic research spending, as one potential path to reducing the future debt-to-GDP ratio.

Development Drains U.S. Economic Productivity, Relative to Other Research Types

We also find a negative association between development spending and productivity. Development represents the largest category of federal R&D historically; this finding suggests that shifting marginal dollars away from development would strictly improve projected future productivity outcomes. This finding suggests a misallocation of R&D spending against the objective of economic productivity.

Several mechanisms could generate this pattern. First, federal development spending is heavily concentrated in defense and space missions, and the productivity returns to this type of work may flow only weakly to the economic productivity measures captured in this analysis. Our productivity data covers the U.S. private business sector and uses cost-based deflators for federal output. Productivity gains realized inside government operations or in defense contractor production of military goods are therefore likely understated relative to the gains realized in private commercial production. Second, the technologies developed for specific military applications or space vehicles may have few uses outside those narrow domains and heightened information security that limits dual-use applications.

Third, to the extent that federal development competes for scientific talent and resources with more productivity-enhancing activities, it could reduce aggregate productivity. And fourth, development spending may substitute for adoption of existing technologies, which would raise measured productivity in the short run but would not be captured in our R&D measures. If development crowds out technology adoption, measured productivity could decline even as the underlying technological capability improves. Each of these mechanisms is consistent with our findings, though they imply different normative interpretations.

The primary implication for policymakers, thus, is to explicitly consider the objectives of R&D policy and whether existing development investments are aligned with public priorities. The comparative advantage of Federal R&D dollars, when it comes to productivity, appears to be in basic research. At the same time, there may be investments – such as military technologies— that the private sector otherwise would not produce without Federal investments. More analysis of effective public production systems and conditions for contractor productivity may help close the gap between noneconomic objectives and macroeconomic outcomes; otherwise, tradeoffs could be considered more explicitly in budgeting priorities.

Limitations and Future Directions: Statistical Power, Causal Inference, and Spillovers

Several limitations of our analysis suggest directions for future research. Due to aggregation, our analysis relies on a small number of data points, limiting statistical power and estimation

precision. Further, our use of aggregate federal R&D data masks substantial heterogeneity across agencies, programs, and scientific fields. Other work (e.g., Fieldhouse & Mertens 2024, Moretti et. al. 2025) has suggested that productivity effects differ between defense R&D and nondefense R&D, and our aggregate estimates are not designed to capture this heterogeneity. Future work with more disaggregated data could examine whether the compositional patterns we document hold within specific domains or reflect differences across domains, by, for example, decomposing basic research into different subcomponents or categories.

Our time-series identification relies on variation that may be confounded by unobserved factors correlated with both R&D composition and productivity. Alternative identification strategies, such as instrumental variables based on political or institutional factors that affect R&D composition independently of productivity considerations, could provide complementary evidence on causal effects.

The international dimension is also absent from our analysis, though there is some evidence of cross-border spillovers from public R&D investment (e.g., as documented by Moretti et. al. 2025 in the defense sector). Other countries have made different compositional choices in their public R&D investments, and cross-country comparison could provide additional identifying variation. However, institutional differences across countries complicate direct comparison and would require careful attention to context.

Conclusion: Opportunity for Compositional Shifts in U.S. R&D Spending

This paper has examined the relationship between the composition of federal R&D spending and U.S. productivity growth using data spanning 67 years. Our analysis finds substantial heterogeneity in the estimated productivity associations across R&D types. Shocks that increase the share of basic research in the federal portfolio are associated with cumulative productivity gains of approximately 2.1 percentage points over ten years. Applied research and R&D plant investments also show positive associations of 1.7 and 1.4 percentage points, respectively. In contrast, positive shocks to the development share are associated with productivity declines of approximately 2.2 percentage points.

These compositional effects substantially exceed the aggregate R&D effects we can detect, suggesting that how federal R&D is allocated may matter more for productivity than the overall level of investment. Portfolio optimization analysis indicates that historical allocations have been tilted toward development relative to compositions that would maximize expected productivity, implying potential gains from reallocation toward basic research and research infrastructure.

Our identification strategy cannot definitively establish causality. The constructed shocks represent unpredictable changes in R&D composition, but unpredictability does not guarantee exogeneity with respect to productivity outcomes. Diagnostic tests support the interpretation of our constructed shocks as valid, but there remains the possibility of unobserved confounding

factors. We interpret our results as informative associations that motivate further investigation rather than definitive causal estimates.

Subject to these limitations, our findings argue that science policy deliberations should attend to the composition of federal R&D investments alongside the overall level of R&D spending. The productivity consequences of compositional choices are large enough to warrant weight alongside national security, human capital development, public health, and other objectives of federal R&D. We caution that our estimates reflect responses to shocks of standard-deviation magnitude; large reallocations could generate different returns through general equilibrium effects that our linear framework does not capture. Further research using alternative identification strategies and more disaggregated data will strengthen the evidentiary foundation for compositional policy recommendations.

Appendix A

This appendix provides additional detail on data construction, estimation procedures, diagnostic tests, and supplementary results.

Properties of Constructed Shocks

The validity of our approach depends on the constructed shocks representing genuinely unpredictable changes rather than forecastable variation or systematic responses to omitted factors. We conduct two types of diagnostic tests.

First, we examine whether the shocks exhibit serial correlation, which would suggest predictable components. Table A.1 reports Ljung-Box and Breusch-Godfrey tests for the absence of serial correlation.

Table A.1. Serial Correlation Tests for Shocks

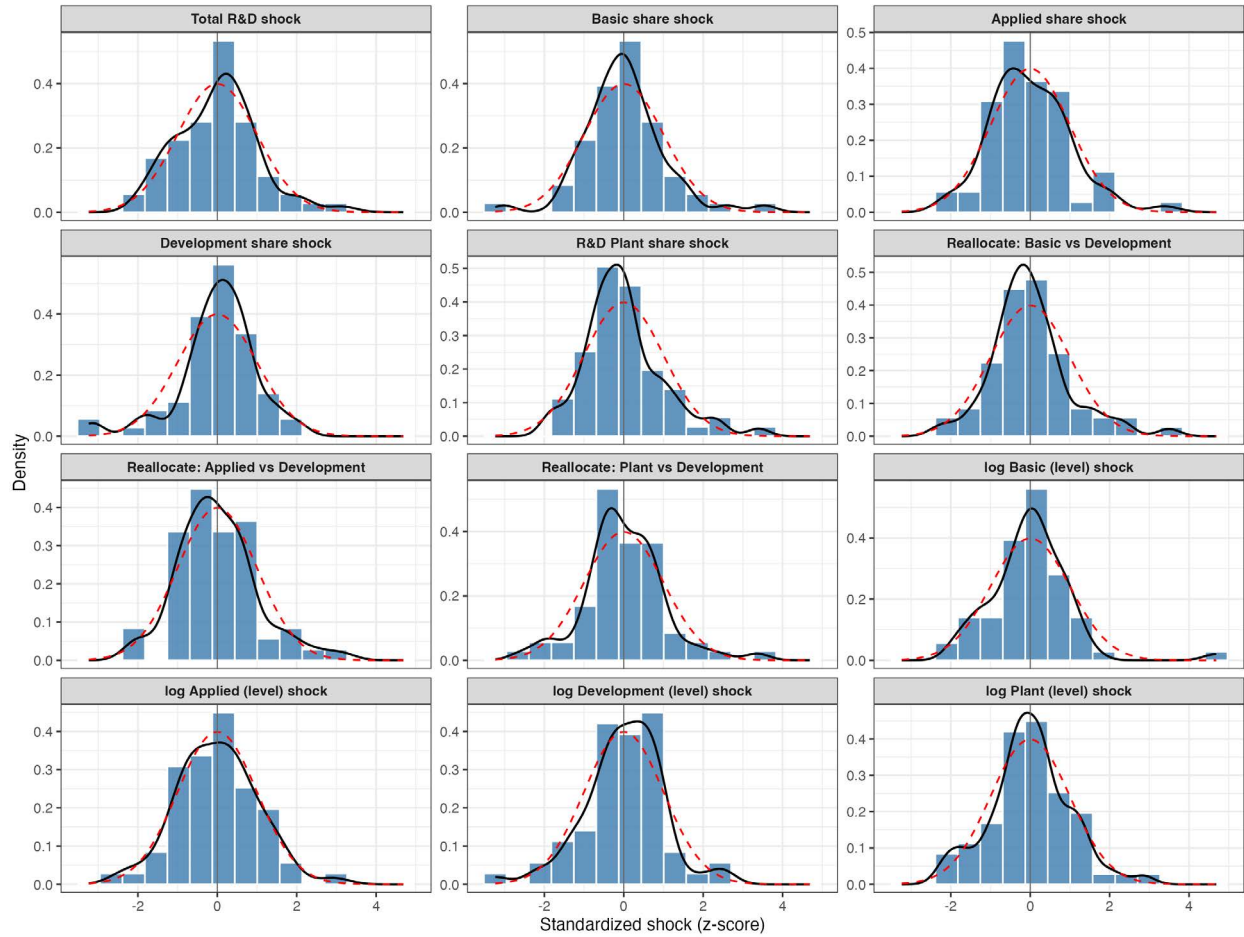
	Ljung-Box χ^2 (df=6)	p-value	Breusch-Godfrey χ^2 (df=4)	p-value
Total	6.14	0.41	2.31	0.68
Basic Share	2.34	0.89	1.36	0.85
Applied Share	6.12	0.41	1.49	0.83
Development Share	6.29	0.39	1.32	0.86
Plant Share	1.09	0.98	0.46	0.98

Notes: Ljung-Box test examines autocorrelation through lag 6. Breusch-Godfrey test examines autocorrelation through lag 4.

All shocks pass both tests at conventional significance levels, with p-values ranging from 0.68 to 0.98. This result indicates that the constructed shocks do not exhibit detectable serial correlation, consistent with their interpretation as unpredictable changes.

We also examine the empirical distribution of the constructed shocks. Figure A.1 presents histograms of each standardized shock series overlaid with the standard normal density.

Figure A.1. Empirical Distributions of Constructed Shocks



Notes: Each panel shows the empirical distribution of the indicated standardized shock series (z-score). The solid black line is a kernel density estimate. The dashed red line is the standard normal density for reference. Shocks are constructed as residuals from forecasting equations with four lags and macroeconomic controls. SOURCE: Authors' calculations from NSF and OMB historical tables.

Alternative Lag Specifications

Our main results use four lags in both the shock construction and the Local Projections estimation. Substantively, four years corresponds to a single presidential term, which represents a natural institutional cycle for federal R&D priority setting. Incoming administrations typically revisit R&D budget priorities as part of a broader policy agenda, and the federal budgeting process introduces additional implementation lags that extend the influence of any given administration's priorities across its full term. A four-year lag structure thus allows the forecasting equations to condition on a complete cycle of political priority setting, helping to absorb the predictable component of compositional change driven by shifts in administrations.

However, we also evaluate whether the primary results are sensitive to lag structure. Table A.2 examines robustness to alternative lag lengths.

Table A.2. Cumulative LP Effects Under Alternative Lag Specifications

Innovation	2 Lags	4 Lags (Baseline)	6 Lags
Basic Share	2.15** (0.90)	2.09** (0.93)	1.10 (0.97)
Applied Share	1.68* (0.91)	1.70* (0.92)	1.05 (0.95)
Development Share	-2.26** (0.94)	-2.23** (0.93)	-1.56 (0.96)
Plant Share	1.37* (0.83)	1.38 (0.87)	2.01** (0.92)

Notes: Cumulative labor productivity effects at $h=10$ under alternative lag specifications for shock construction and LP estimation.

The estimates are qualitatively and quantitatively stable for shorter lags, but for longer lags the results attenuate and generally become more imprecisely estimated. The attenuation at longer lags suggests the relationship becomes more difficult to detect as the specification becomes more demanding of the data.

Stability Across a Structural Break

To examine whether our estimated associations are stable over time, we re-estimate the models allowing for a structural break at the year 1980. We selected this year due to the Bayh-Dole Act of 1980, which changed incentive structures around research by allowing U.S. universities and academic research to own patents on their research.

Figure A.2 summarizes the results of the structural break analysis, showing the cumulative difference in impulse responses between the post-1980 and pre-1980 periods (i.e., post-1980 estimated effect, minus pre-1980 estimated effect) for each shock-component pair. Filled dots denote horizons where the difference is statistically significant at the 5-percent level; shaded bands reflect 95-percent bootstrap confidence intervals.

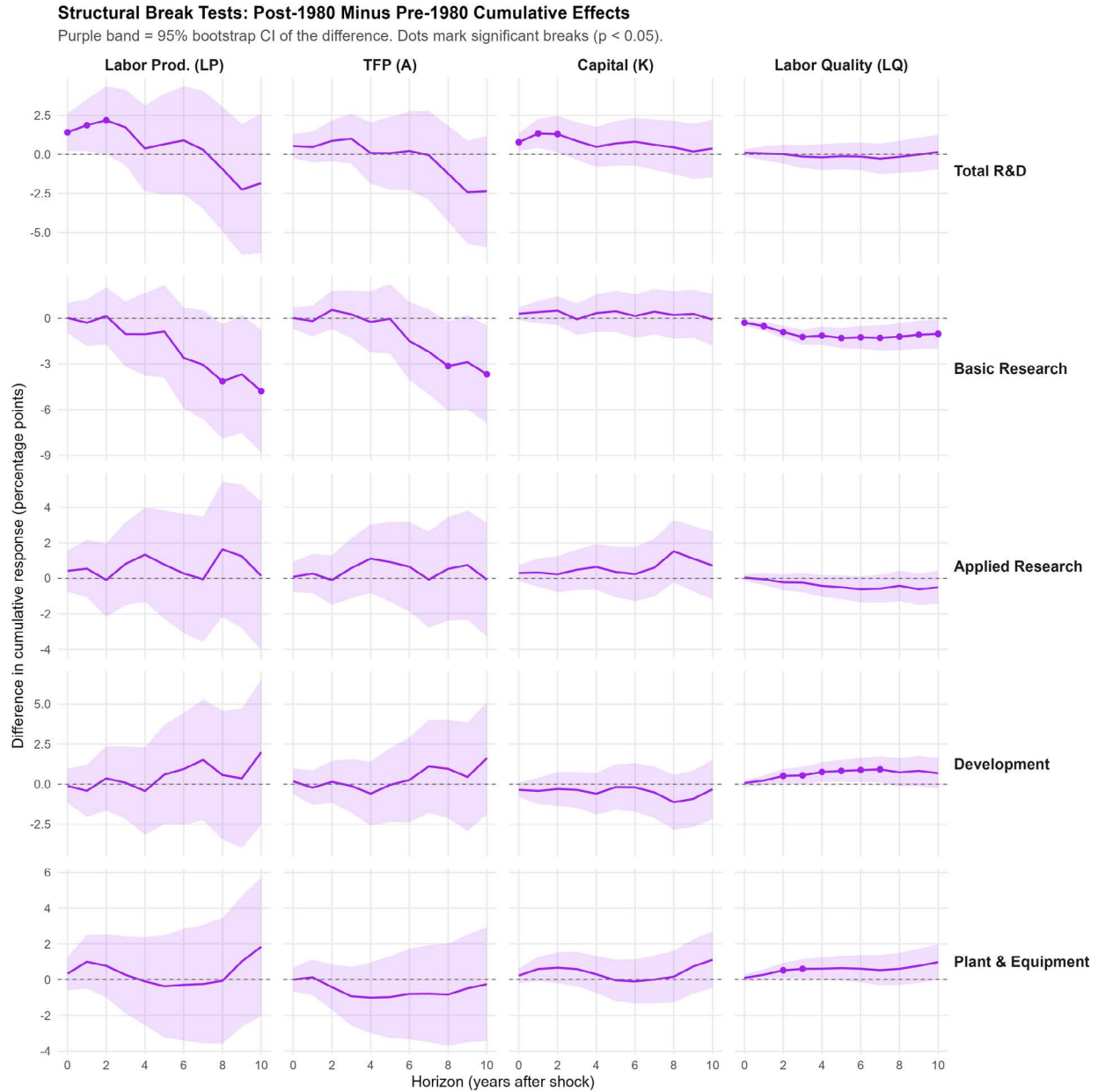
The most consistent finding is stability for our primary measure of innovation. We find there is no significant difference between periods in the relationship between any category of R&D spending and TFP, our primary measure of innovation. This holds across all five shock types and all horizons. More broadly, the majority of shock-component pairs show no evidence of a structural break.

There are two exceptions in the short-run response to total R&D shocks. The cumulative effects on labor productivity and capital deepening differ significantly across periods at horizons zero to two, driven by a sign reversal in the capital response (negative pre-1980, positive post-1980), as shown in Figure A.3. These differences are only marginally significant and converge toward zero by horizon 3 (as shown in Table A.3), which suggests there is no permanent shift in

the R&D-productivity relationship. Confidence intervals at longer horizons are wide, consistent with the limited sample size available for cumulative estimation at extended horizons as well as greater likelihood of confounding factors.

The most notable structural break appears in the relationship between basic research spending and hours worked, where the difference is significant at every horizon ($p < 0.01$). This difference (depicted in Figure A.4) reflects the disappearance of a previously strong positive effect: pre-1980, basic research shocks were associated with cumulative increases in hours worked of 0.33 to 1.47 percentage points; post-1980, the corresponding estimates are indistinguishable from zero. Importantly, no parallel break appears in the TFP or capital responses to basic research, suggesting that the pre-1980 hours effect may have reflected extensive-margin labor absorption rather than productivity-enhancing reallocation. The absence of this channel after 1980 is consistent with a shift toward more capital-or knowledge-intensive modes of translating basic research into economic activity, possibly facilitated by the commercialization incentives introduced by the Bayh-Dole Act.

Figure A.2. Structural Break: Cumulative Differences Over Time

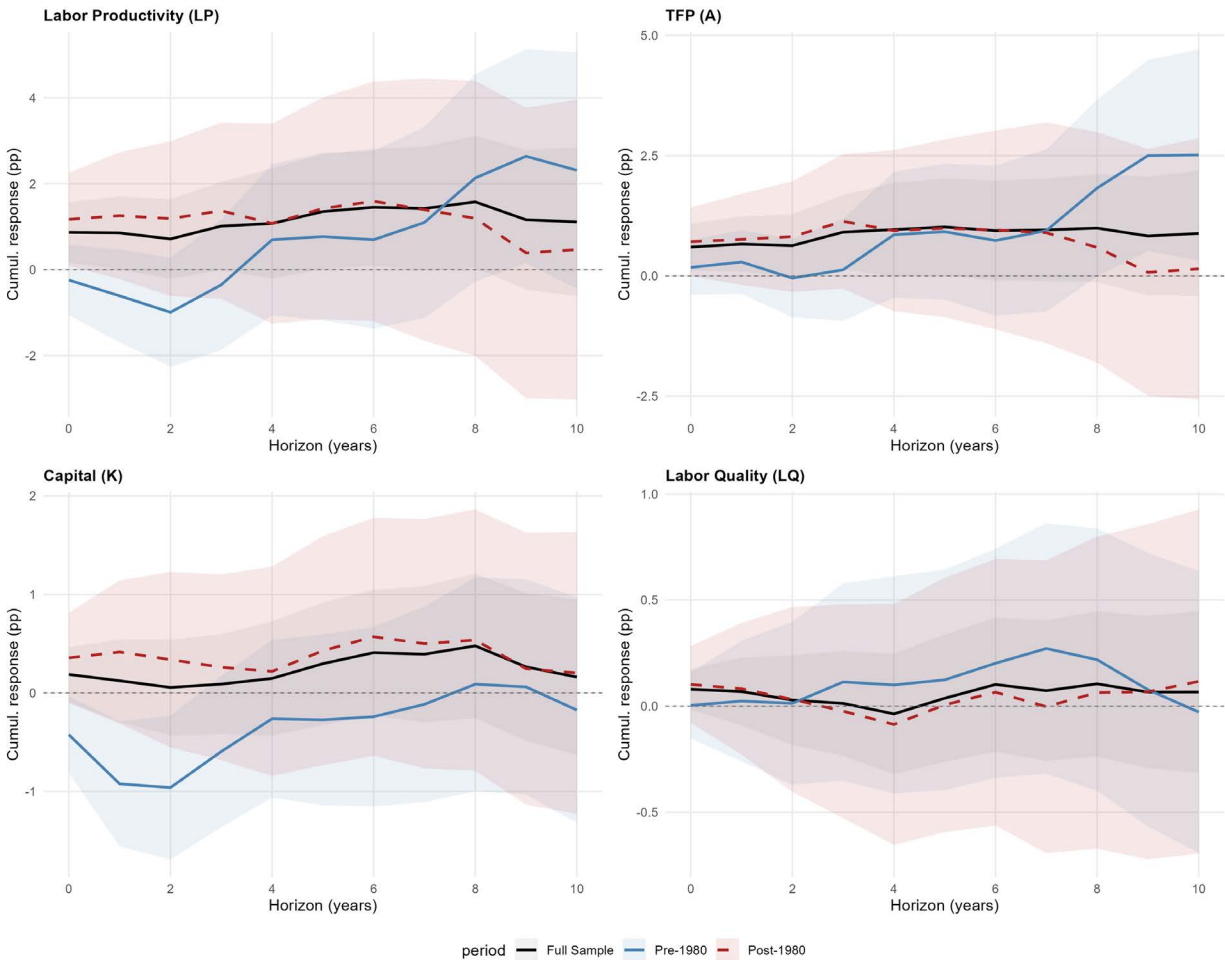


Notes: This figure shows differences in estimated coefficients for cumulative effects (post-1980 specifications minus the pre-1980 specifications), including controls for the output gap (CBO data), federal funds rate (FRED data), and total R&D spending level (NSF data). Labor productivity is defined as the sum of coefficients on TFP (A), capital (K), and labor quality (LQ). The shaded areas correspond to a 95-percent confidence interval for the differences in each time period, using bootstrap standard errors (400 draws, moving block bootstrap, block length = 5). Significant differences are marked with solid dots.

Figure A.3. Structural Break Overlay: Total R&D, Cumulative Response Over Time

Structural Break Overlay: Total R&D

Black = full sample | Blue = pre-1980 | Red dashed = post-1980 (95% CI bands)

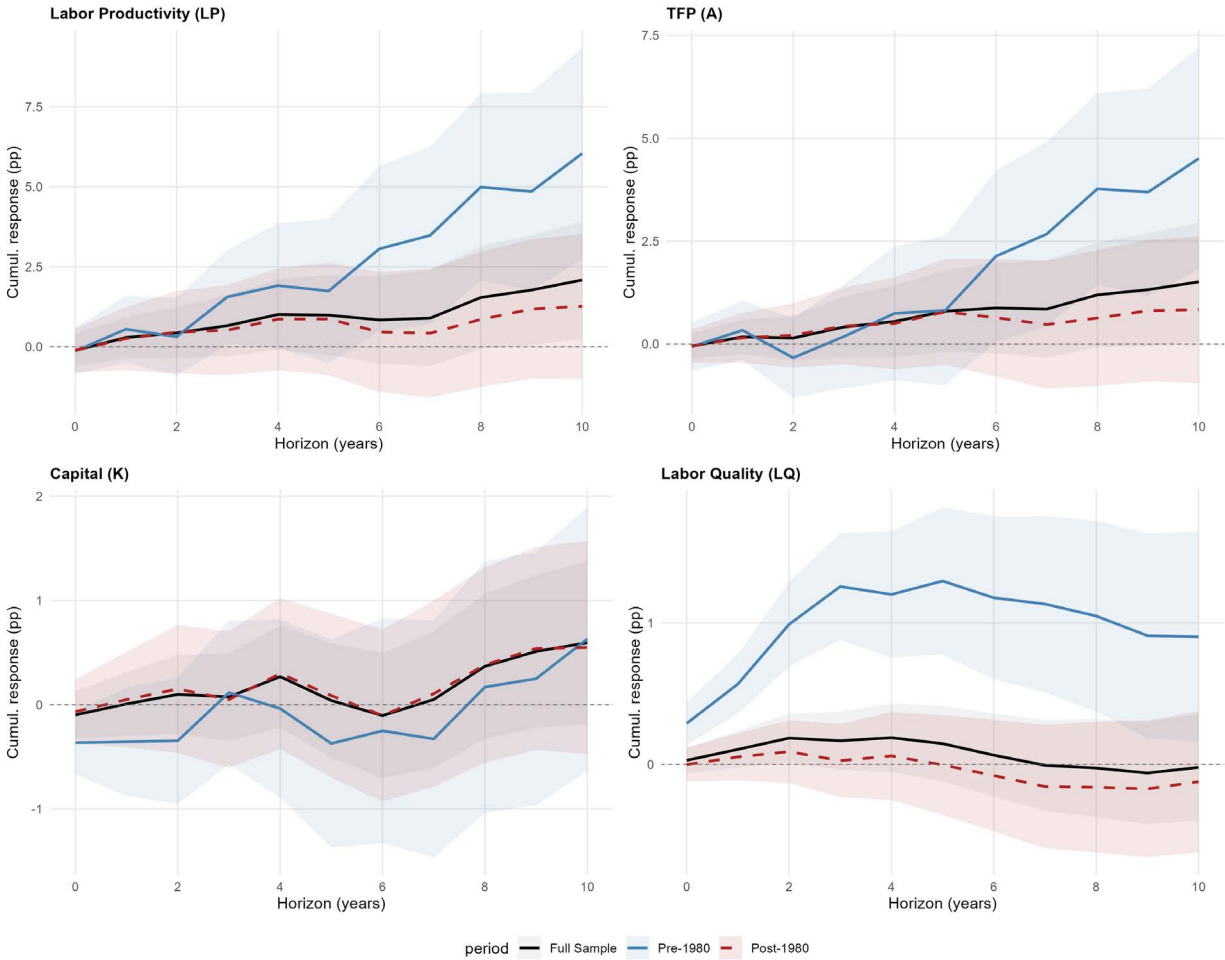


Notes: Estimates are the coefficients on cumulative response of each outcome to unexplained variation in total R&D spending, including controls for the output gap (CBO data), federal funds rate (FRED data), and total R&D spending level (NSF data). Labor productivity is defined as the sum of coefficients on TFP (A), capital (K), and labor quality (LQ). The shaded areas correspond to a 95-percent confidence interval for the coefficients in each time period, using bootstrap standard errors (400 draws, moving block bootstrap, block length = 5).

Figure A.4. Structural Break Overlay: Basic Research, Cumulative Response Over Time

Structural Break Overlay: Basic Research

Black = full sample | Blue = pre-1980 | Red dashed = post-1980 (95% CI bands)



Notes: Estimates are the coefficients on cumulative response of each outcome to unexplained variation in basic R&D, including controls for the output gap (CBO), federal funds rate (FRED), and total R&D spending level (NSF). Labor productivity is defined as the sum of coefficients on TFP (A), capital (K), and labor quality (LQ). The shaded area corresponds to a 95-percent confidence interval using bootstrap standard errors (400 draws, moving block bootstrap, block length = 5).

Table A.3. Structural Break Differences

Shock	Component	At h	Peak diff.	(SE)	Sig. horizons (5%)
Total R&D	Labor Prod. (LP)	0	1.414	-0.607	3 / 11
	TFP (A)	9	-2.427	-1.660	0 / 11
	Capital (K)	1	1.339	-0.489	3 / 11
	Labor Quality (LQ)	0	0.099	-0.111	0 / 11
Basic Research	Labor Prod. (LP)	10	-4.334	-2.130	2 / 11
	TFP (A)	8	-2.412	-1.414	0 / 11
	Capital (K)	3	-0.470	-0.531	0 / 11
	Labor Quality (LQ)	3	-1.389	-0.303	11 / 11
Applied Research	Labor Prod. (LP)	8	2.217	-1.847	0 / 11
	TFP (A)	4	1.091	-0.917	0 / 11
	Capital (K)	8	1.492	-0.871	0 / 11
	Labor Quality (LQ)	6	-0.548	-0.408	0 / 11
Development	Labor Prod. (LP)	2	0.595	-0.922	0 / 11
	TFP (A)	4	-0.869	-0.933	0 / 11
	Capital (K)	8	-1.162	-0.863	0 / 11
	Labor Quality (LQ)	5	0.783	-0.377	3 / 11
Plant & Equipment	Labor Prod. (LP)	1	1.320	-0.797	0 / 11
	TFP (A)	4	-0.817	-1.050	0 / 11
	Capital (K)	1	0.770	-0.392	1 / 11
	Labor Quality (LQ)	2	0.502	-0.241	1 / 11

Notes: Peak diff. reports the largest cumulative difference (post-1980 minus pre-1980) by absolute t-statistic across horizons $hh = 0, \dots, 10$. Standard errors correspond to the peak difference horizon. Sig. horizons reports the fraction of horizons with $p < 0.05$. Bootstrap standard errors (400 draws, moving block bootstrap, block length = 5).

Lead Tests

A standard concern in time-series designs that exploit policy changes is that the measured shock may still proxy for slow-moving and forward-looking policy adjustments or other latent forces that also predict future productivity. To evaluate this potential concern, we implement a lead test that estimates whether future values of the constructed R&D composition shocks have incremental predictive power for labor productivity, conditional on the current shock and standard controls. For each composition shock, we estimate regressions that include three-year

leads of the shock, alongside the contemporaneous shock and controls. We then test the joint null hypothesis that the coefficient on these leads are jointly zero.²⁶

Table A.4 reports the joint lead test for each composition shock and horizon. Across the four composition shocks and three horizons evaluated, we do not find strong evidence that future shocks are predictive of the outcome, conditional on the controls, though the joint test for the Plant and basic share are marginally significant at the ten-year horizon. The absence of systematic predictability from future shocks is consistent with the intended interpretation of the composition shocks as shock-like reallocations, rather than mechanical reflections of pre-existing productivity trends.

Table A.4. Lead Tests (F-stat and p-values for Joint Significance)

Innovation	h=0	h=5	h=10
Basic Share	1.25 (0.277)	0.70 (0.550)	2.09* (0.083)
Applied Share	0.95 (0.433)	0.30 (0.883)	1.30 (0.250)
Development Share	0.94 (0.363)	0.31 (0.830)	1.10 (0.340)
Plant Share	0.91 (0.393)	1.48 (0.210)	2.28* (0.080)

Notes: P-values from F-tests of joint significance of share leads in LP regressions. Low p-values suggest predictability of the productivity outcome by future shocks.

Placebo Tests

Are the estimated productivity responses in Table 3 tied to the historical timing of R&D share reallocations, or would similarly large effects appear even if the same shock sequence occurred in different years? To evaluate this, we implement a circular-shift placebo test. For each R&D share shock, we preserve the entire realized shock sequence (including serial correlation and low frequency structure) but rotate it by a random offset in calendar time and re-estimate the same LP specification. This test generates a demanding placebo null that the shock retains realistic time series properties, but its timing is no longer aligned with the historical sequency of productivity movements and policy episodes. Small placebo p-values indicate that our baseline cumulative effects are unusually large relative to estimates obtained under alternative calendar

²⁶ We implement this using a moving block bootstrap with the same block length as our other analysis (L=5 years). In each bootstrap replication, we resample blocks of the raw annual data, rebuild the full set of shocks inside the draw using the same shock-construction procedure as in the main analysis, re-estimate the lead regression, and record the vector of lead coefficients. We then construct a joint Wald statistic using the bootstrap covariance matrix of the lead coefficients. Table A.4 reports the bootstrap p-values computed under the recentered null (i.e., recentering the bootstrap distribution so that the mean lead vector equals zero). In all specifications, we use 300 bootstrap replications.

alignments, suggesting that the results are not driven primarily by generic persistence or comovements.

Table A.5 presents the results from this exercise. The placebo p-values for the applied and development share results are small, indicating that the estimated effects are unusual relative to the distribution under randomly shifted timing and are therefore meaningfully tied to the historical timing of actual reallocations. The basic share result is directionally consistent with this pattern across most channels but is less precisely distinguished from the placebo distribution. The plant share placebo p-values are large, which is consistent with the imprecisely estimated plant share effects in our primary results.

Table A.5. Placebo Test

Shock (share spec)	TFP (A)	Capital deepening (K)	Labor Quality (LQ)	Labor productivity (LP)
Basic share shock	1.51* (0.083)	0.60* (0.091)	-0.02 (0.914)	2.09** (0.036)
Applied share shock	1.10* (0.079)	0.86** (0.010)	-0.25 (0.592)	1.70** (0.023)
Development share shock	-1.36** (0.045)	-0.92** (0.018)	0.04 (0.836)	-2.23** (0.028)
Plant share shock	0.96 (0.440)	0.77*** (0.000)	-0.35* (0.055)	1.38 (0.304)

Notes: Entries report the cumulative (ten-year) LP response for each outcome to a one standard deviation innovation in the indicated R&D share. Placebo-based p-values in parentheses.

Abbreviations

R&D	Research and development
TFP	Total factor productivity
LP	Linear projections
GDP	Gross domestic product
NIH	National Institute of Health
NCSES	National Center for Science and Engineering Statistics
NSF	National Science Foundation
OMB	Office of Management and Budget

References

- Akcigit, U., Hanley, D., & Serrano-Velarde, N. (2021). Back to basics: Basic research spillovers, innovation policy, and growth. *Review of Economic Studies*, 88(1), 1-43.
- Azoulay, P., Graff Zivin, J., Li, D., & Sampat, B. N. (2019). Public R&D investments and private-sector patenting: Evidence from NIH funding rules. *Review of Economic Studies*, 86(1), 117-152.
- Bloom, N., Schankerman, M., & Van Reenen, J. (2013). Identifying technology spillovers and product market rivalry. *Econometrica*, 81(4), 1347-1393.
- Deller, Steven and Tessa Conroy, (2014). Innovation and Economic Performance: R&D within Wisconsin. *Patterns of Economic Growth and Development*. Study Series No. 5
- Fernald, J. G. (2014). A quarterly, utilization-adjusted series on total factor productivity. *Federal Reserve Bank of San Francisco Working Paper 2012-19*.
- Fleming, L., & Sorenson, O. (2004). Science as a map in technological search. *Strategic Management Journal*, 25(8-9), 909-928.
- Griliches, Z. (1979). Issues in assessing the contribution of research and development to productivity growth. *Bell Journal of Economics*, 10(1), 92-116.
- Griliches, Z. (1992). The search for R&D spillovers. *Scandinavian Journal of Economics*, 94(Supplement), 29-47.
- Guellec, D., & Van Pottelsberghe de la Potterie, B. (2004). From R&D to productivity growth: Do the institutional settings and the source of funds of R&D matter? *Oxford Bulletin of Economics and Statistics*, 66(3), 353-378.
- Gullo, Theresa, et. al.. "Estimating the Economic and Budgetary Effects of Research Investments." *Tax Policy and the Economy* 39.1 (2025): 53-81.
- Hall, B. H., Mairesse, J., & Mohnen, P. (2010). Measuring the returns to R&D. In B. H. Hall & N. Rosenberg (Eds.), *Handbook of the Economics of Innovation* (Vol. 2, pp. 1033-1082). Elsevier.
- Jentsch, C., & Lunsford, K. G. (2019). The dynamic effects of personal and corporate income tax changes in the United States: Comment. *American Economic Review*, 109(7), 2655-2678.
- Jones, C. I., & Williams, J. C. (1998). Measuring the social return to R&D. *Quarterly Journal of Economics*, 113(4), 1119-1135.

- Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1), 161-182.
- Jorgenson, D. W., Gollop, F. M., & Fraumeni, B. M. (1987). *Productivity and U.S. Economic Growth*. Harvard University Press.
- Mansfield, E. (1980). Basic research and productivity increase in manufacturing. *American Economic Review*, 70(5), 863-873.
- Moretti, E., Steinwender, C., & Van Reenen, J. (2019). The intellectual spoils of war? Defense R&D, productivity and international spillovers. *NBER Working Paper No. 26483*.
- Moris, F. and Pece, C. (2022). Definitions of Research and Development: An Annotated Compilation of Official Sources, *National Science Foundation, National Center for Science and Engineering Statistics (NCSES)*, NCSES 22-209. As of May 29, 2026:
<https://ncses.nsf.gov/pubs/ncses22209>
<https://wayback.archive-it.org/5902/20160210164701/http://www.nsf.gov/statistics/randdef/fedgov.cfm>
- Ramey, V. A. (2016). Macroeconomic shocks and their propagation. In J. B. Taylor & H. Uhlig (Eds.), *Handbook of Macroeconomics* (Vol. 2, pp. 71-162). Elsevier.
- Solow, R. M. (1957). Technical change and the aggregate production function. *Review of Economics and Statistics*, 39(3), 312-320.

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